

THE ANATOMY OF A PHONE NUMBER

The Architecture of a Phone Number

VOLUME I OF III

Identifiers, numbering plans and 150 years of switching

Why a telephone number is an identifier rather than an address, how the international numbering plan is organised, what the SIM, the handset and the subscription each contribute, and how 150 years of switching technology changed underneath a string of digits that barely changed at all.



A LAYERED STUDY OF TELEPHONE NUMBERING, MOBILE NETWORKS AND PACKET TELEPHONY

PART I / III

About this volume

Why a telephone number is an identifier rather than an address, how the international numbering plan is organised, what the SIM, the handset and the subscription each contribute, and how 150 years of switching technology changed underneath a string of digits that barely changed at all.

This is one of three volumes. Each descends a level further from the digits a human types toward the electromagnetic wave that eventually carries the speech. The volumes are meant to be read in order, but each is self-contained.

VOLUME I

The Architecture of a Phone Number

VOLUME II

Anatomy of a 5G Voice Call

VOLUME III

Packet-Level Telecommunications

Diagrams are numbered by chapter. A list of figures follows the contents, so any diagram can be found directly.

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User-facing identity → Location / mobility → IMEI / modem / SIM / radio hardware

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Introduction

A phone number looks like a tiny string of digits, but underneath it sits on top of roughly 150 years of telecommunications engineering, regulatory allocation, switching theory, signaling protocols, databases, radio networks, and increasingly IP-based routing.

The important conceptual correction is this:

A phone number is not the address of your phone. It is a globally managed identifier that the telephone network uses to discover how to reach your subscription/service.

Your handset has other identities. The SIM has identities. The radio network has identities. The IP network has identities. Your phone number is only one layer of the stack.

Let's peel the whole thing apart.

What a Phone Number Actually Is

1.1 The fundamental abstraction

1.2 First distinction: identifier vs address

1.1 The fundamental abstraction

Suppose your Indian mobile number is:

+91 98XX YYYYYY

At the human level:

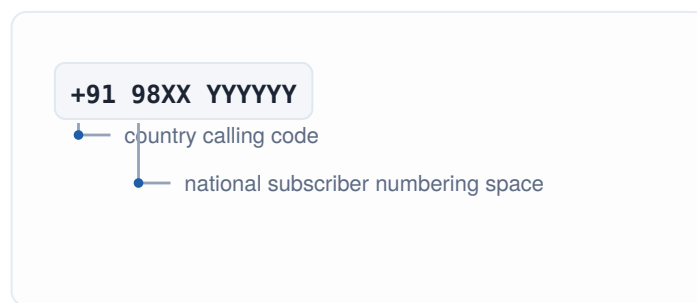


Figure 1.1 Structure of +91 98XX YYYYYY

But the actual telecommunications system does not simply think:

"+9198XXYYYYYY" → radio tower

It is more like:

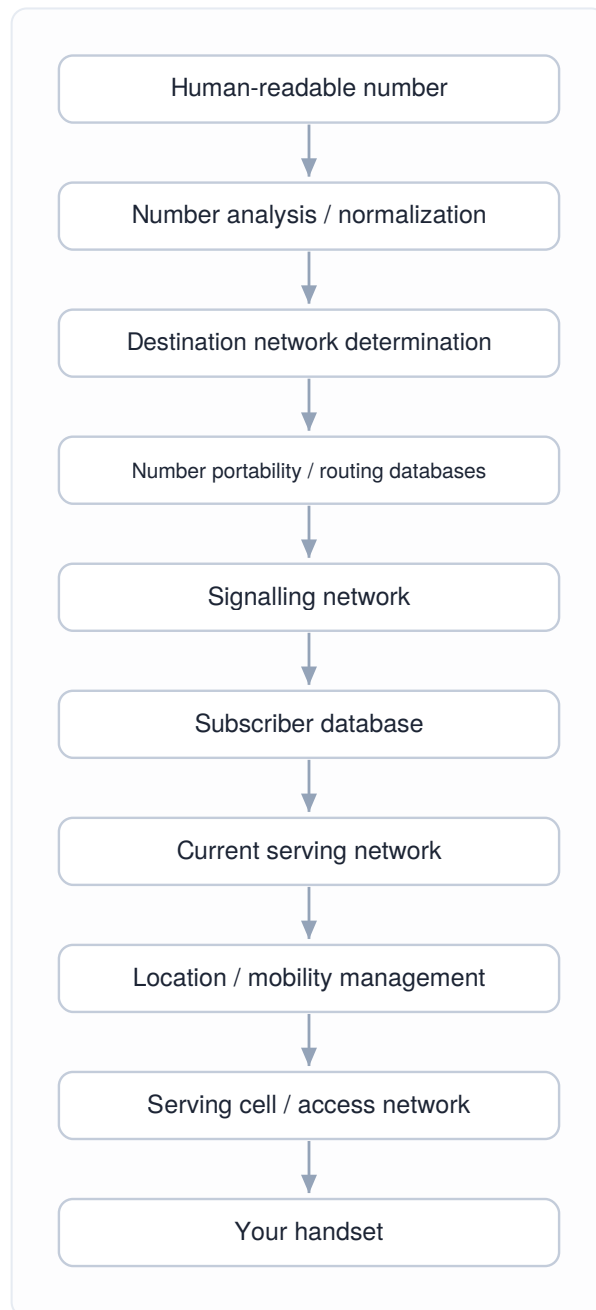


Figure 1.2 Human-readable number → Subscriber database → Your handset

And that architecture has changed dramatically over time.

1.2 First distinction: identifier vs address

This is the most important concept.

In computer networking, you already know:

- MAC address
- IP address
- DNS name
- socket

These aren't the same thing.

Telephony has the same separation.

A simplified model is:

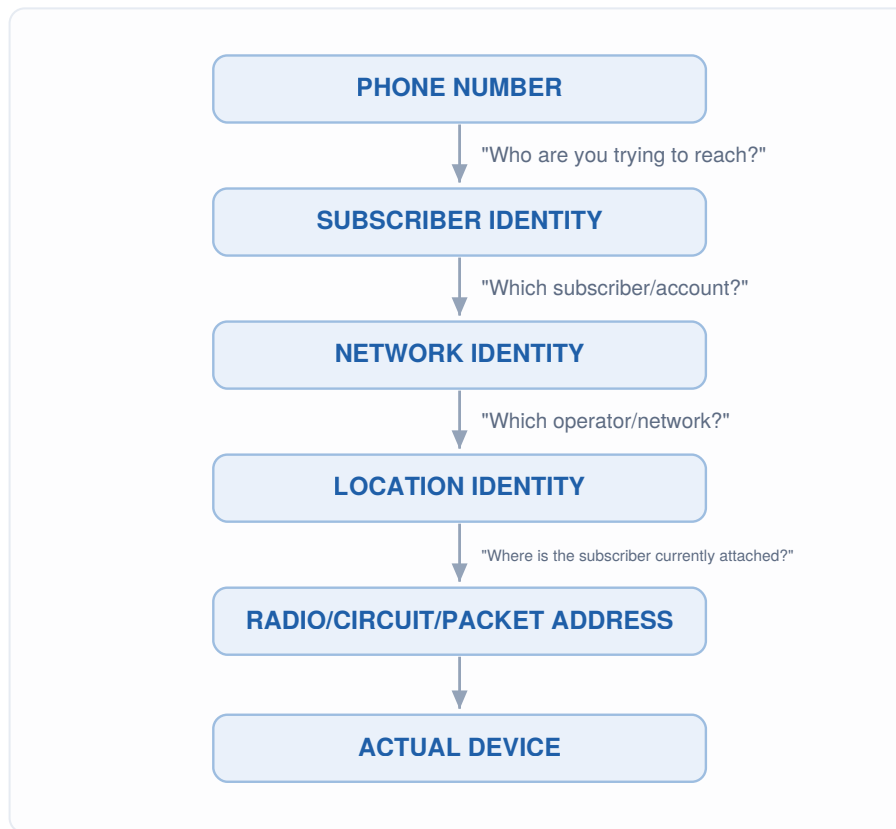


Figure 1.3 Phone number → Location identity → Actual device

So a phone number is closer to a stable logical identifier than a physical network address.

This is why you can change:

- SIM
- handset
- cell tower
- location
- operator

while retaining the same telephone number.

Number portability makes this even more obvious. The FCC describes number portability as allowing customers to keep their number when changing providers, and India implemented nationwide mobile number portability in January 2011. [FCC Complaints](#)

CHAPTER 2

The Global Numbering Plan

2.1 The international structure

2.3 India's number

2.2 Country codes aren't random

2.1 The international structure

The international numbering system is standardized primarily through ITU-T Recommendation E.164.

The current E.164 revision is from February 2026. It defines international public telecommunications numbering structures and currently covers several categories including geographic areas, global services, satellite services, networks, groups of countries, trial resources, and IoT/M2M applications. [ITU](#)

For an ordinary geographic telephone number, conceptually:

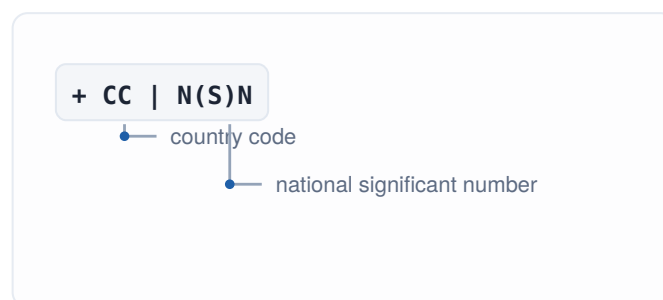


Figure 2.1 Structure of + CC | N(S)N

Where:

CC = Country Code

NSN = National Significant Number

Historically, E.164 has generally operated within a maximum of 15 digits for an international number.

For example:

+91 9876543210

Break it down:

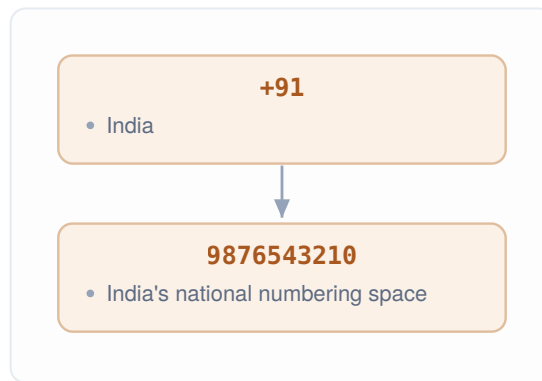


Figure 2.2 +91 → 9876543210

The + isn't actually transmitted as a digit through the telephone network. It is a notation meaning:

"Use the international access procedure appropriate for this country."

For example, depending on where you're dialing from:

+91 9876543210

might be entered using an international dialing prefix such as:

00 91 9876543210

The actual prefix varies by originating country/network.

2.2 Country codes aren't random

The ITU allocates international numbering resources.

Historically, the world was divided into numbering zones.

Very roughly:

- Zone 1** → North America
- Zone 2** → Africa
- Zone 3/4** → Europe
- Zone 5** → Latin America
- Zone 6** → Southeast Asia/Oceania
- Zone 7** → Russia/Kazakhstan
- Zone 8** → East Asia
- Zone 9** → West/South/Central Asia + Middle East

Hence:

- India +91
- Pakistan +92
- Afghanistan +93
- Sri Lanka +94

The exact allocation architecture is more complicated than simply "zone + country", because numbering resources evolved historically.

The ITU maintains the authoritative international allocation framework. ^{ITU}

2.3 India's number

For an ordinary Indian mobile number:

+91 XXXXXXXXXX

you normally have:

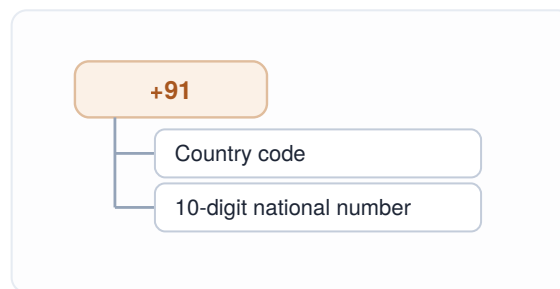


Figure 2.3 +91 and its constituent parts

Example:

+91 9876543210

Historically, Indian mobile numbering ranges were strongly associated with:

- operator
- service area/circle
- subscriber range

But number portability destroyed the simplistic assumption that a prefix tells you the current operator.

For example, you could originally receive:

98XX → Operator A

and later port:

98XX → Operator B

while retaining:

98XX YYYYYY

India's MNP framework was introduced initially in Haryana as a pilot in November 2010 and implemented across all telecom service areas in January 2011. (Press Information Bureau)

That little ten-digit number therefore became considerably less informative than people assume.

The Identity Family: MSISDN, IMSI, IMEI, ICCID

3.1 The phone number's "possessions"

3.2 MSISDN

3.3 IMSI

3.4 IMEI

3.5 ICCID

3.1 The phone number's "possessions"

If you're asking what identities exist around a mobile number, this is where things become interesting.

A modern mobile subscription can involve something like:

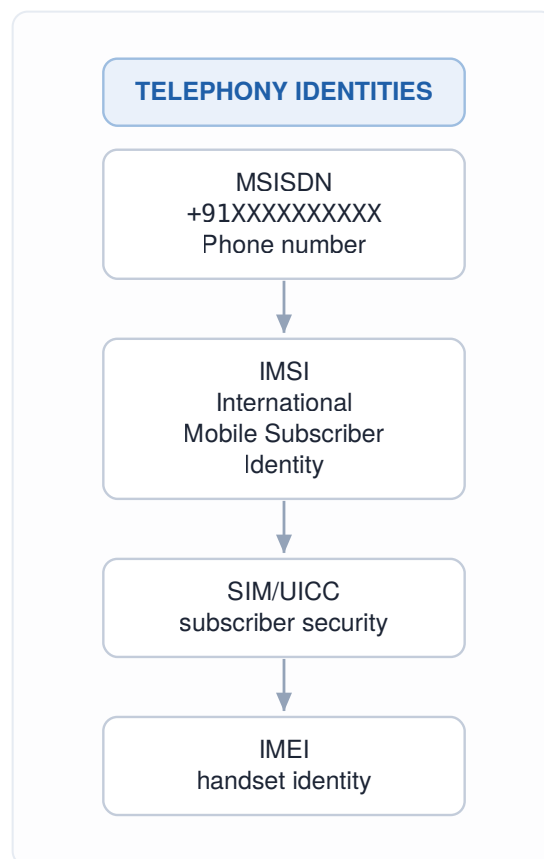


Figure 3.1 Telephony identities → IMSI → IMEI

These are not interchangeable.

3.2 MSISDN

The telephone number associated with a mobile subscriber is generally called the:

MSISDN

Mobile Station ISDN Number.

GSMA defines MSISDN as the structured identifier for the subscriber indicating the A-number. GSMA

Conceptually:

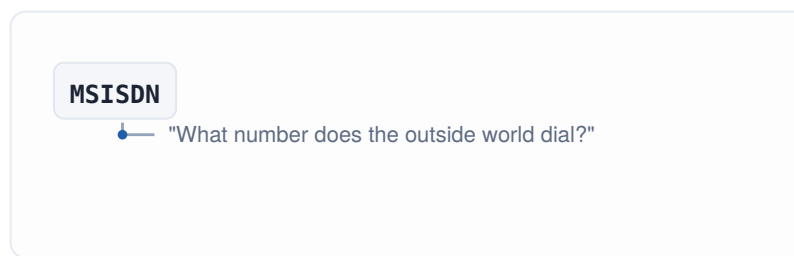


Figure 3.2 Structure of MSISDN

Example:

+919876543210

That's the number your friend knows.

3.3 IMSI

Now we get deeper.

The SIM/subscriber has an:

IMSI

International Mobile Subscriber Identity.

Conceptually:

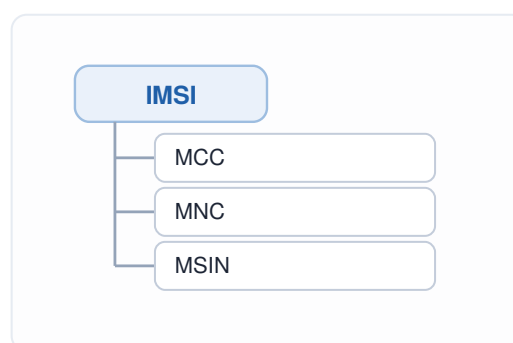


Figure 3.3 IMSI and its constituent parts

Typically:

MCC = Mobile Country Code

MNC = Mobile Network Code

MSIN = Mobile Subscriber Identification Number

The IMSI identifies the subscription within the mobile system.

It is not the same thing as your phone number.

This distinction is fundamental:

- | | |
|--------------------|--------------------------------|
| • MSISDN | • IMSI |
| • telephone number | • mobile subscription identity |

3GPP/ETSI specifications explicitly maintain subscriber information such as IMSI, MSISDN, IMEI and related mobility/security parameters in mobile-network databases. (ETSI)

3.4 IMEI

Then there's the:

IMEI

International Mobile Equipment Identity.

This identifies the equipment, not the subscriber.

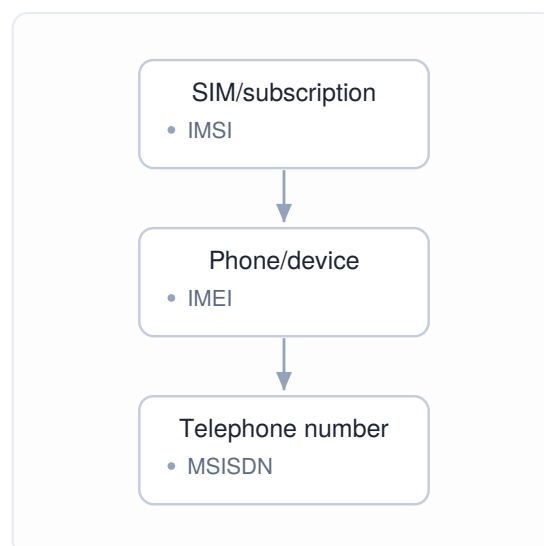


Figure 3.4 SIM/subscription → Telephone number

Therefore:

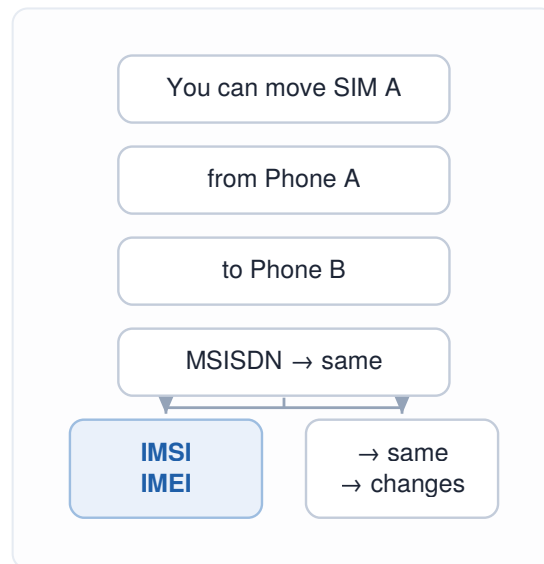


Figure 3.5 You can move SIM A → to Phone B → IMSI / → same

That is the reason the network can distinguish:

"This subscriber"

from:

"This physical handset."

3.5 ICCID

The physical SIM/UICC also has an:

ICCID

Integrated Circuit Card Identifier.

Think:

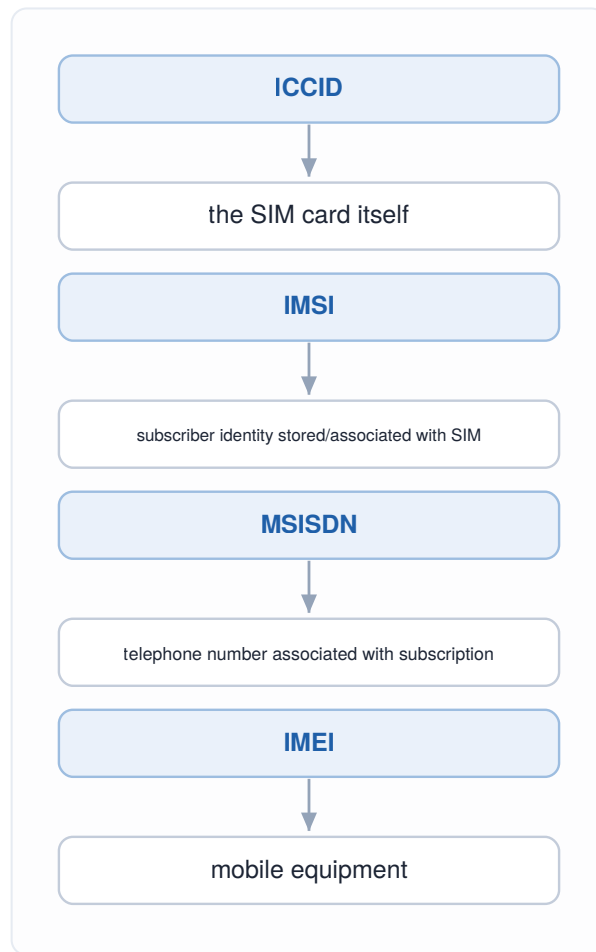


Figure 3.6 ICCID → MSISDN → mobile equipment

Modern eSIMs complicate the physical-card intuition, because the subscription credentials can be provisioned into an embedded secure element rather than a removable plastic SIM.

Human beings invented a card, then removed the card, then embedded the card. Progress, apparently, is making the card invisible.

The Birth of Automatic Switching

4.1 Historical architecture: before automatic dialing

4.2 The Strowger revolution

4.3 The telephone number became an instruction

4.4 Rotary dialing

4.1 Historical architecture: before automatic dialing

Now let's go back to the nineteenth century.

Early telephone networks were extremely primitive by modern standards.

Imagine:

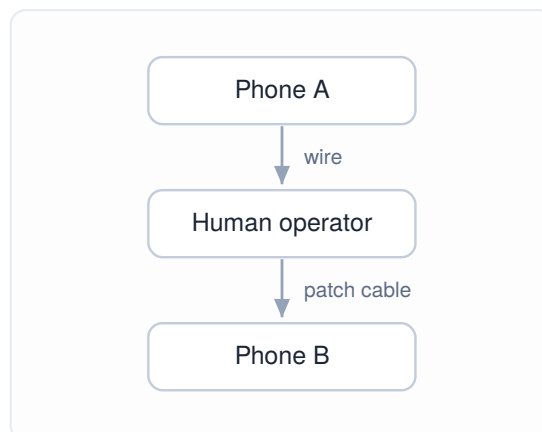


Figure 4.1 Phone A → Phone B

You picked up your telephone and effectively said:

"Connect me to John."

The operator manually connected circuits.

There wasn't necessarily a globally meaningful telephone number in the modern sense.

The network could be organized around:

- subscriber name
- exchange
- local identifier

Telephone directories subsequently became an essential component of this social/network infrastructure. The Library of Congress has surviving telephone directories from the 1890s onward, showing how exchange-based addressing became institutionalized. (Research Guides)

4.2 The Strowger revolution

Then came automatic switching.

Almon Strowger developed an automatic telephone exchange in the late nineteenth century.

The famous motivation is almost absurdly human: Strowger, an undertaker, believed a competing undertaker's wife, who worked as a telephone operator, was redirecting his customers' calls.

So he developed an automated switching system that didn't require her.

The first commercial Strowger exchange began operating in the early 1890s. (IEEE Spectrum)

Architecture:

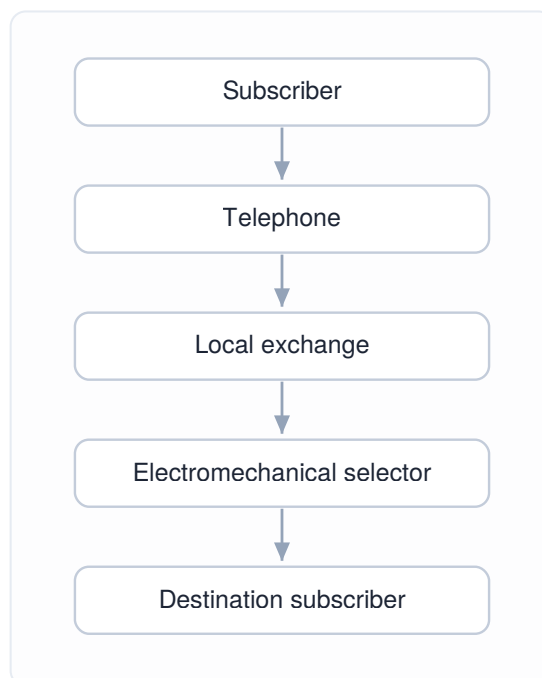


Figure 4.2 Subscriber → Local exchange → Destination subscriber

Now the caller could directly select a destination.

That fundamentally changed what a number meant.

4.3 The telephone number became an instruction

This is a subtle but important idea.

A telephone number isn't merely:

ID = 123456

It was historically closer to:

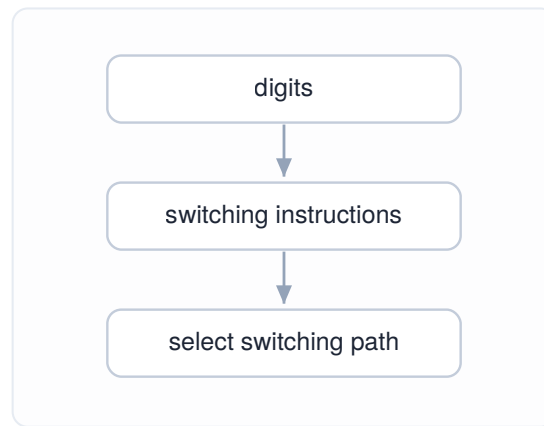


Figure 4.3 digits → select switching path

For example, imagine a primitive exchange:

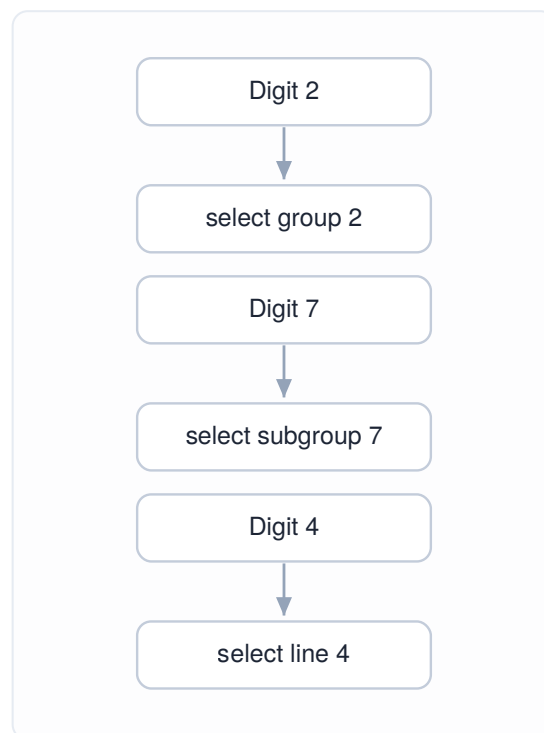


Figure 4.4 Digit 2 → select subgroup 7 → select line 4

The digits were progressively interpreted by switching equipment.

This is digit analysis.

Modern networks still do digit analysis, although the machinery is now software, signaling systems and databases instead of giant electromechanical selectors.

4.4 Rotary dialing

Then came the rotary dial.

The user physically generated pulses corresponding to digits.

Conceptually:

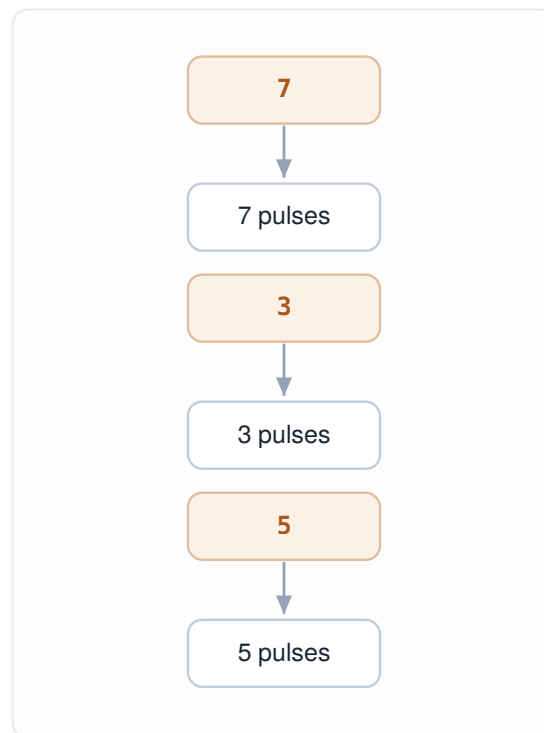


Figure 4.5 7 → 3 pulses → 5 pulses

The exchange counted them.

Later, DTMF replaced pulse dialing for many applications.

DTMF gives us the familiar telephone keypad:

- 1 2 3
- 4 5 6
- 7 8 9
- * 0 #

Each key generates two simultaneous frequencies.

For example:

5
770 Hz + 1336 Hz

The receiver detects the pair.

This became enormously useful for:

- automated menus
- IVR
- banking systems
- voicemail
- remote control systems

From Exchanges to National Networks

5.1 Local exchanges became networks

5.2 Area codes

5.3 But numbering isn't necessarily geography anymore

5.4 Number portability broke the old model

5.1 Local exchanges became networks

One exchange isn't enough for a city.

So the architecture became hierarchical:

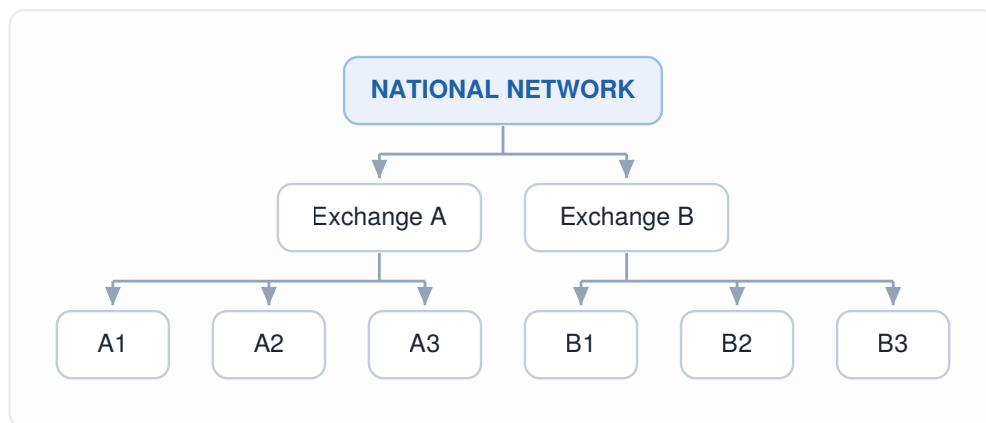


Figure 5.1 National network → A1 / a2 / a3 / b1 / b2 / b3

Eventually:

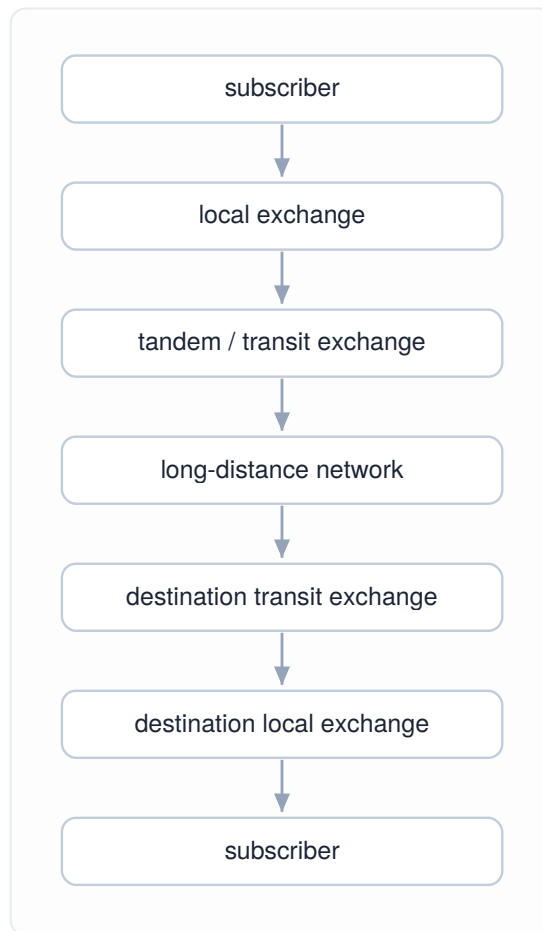


Figure 5.2 subscriber → long-distance network → subscriber

The number structure evolved partly because the network needed enough information to progressively determine where to send the call.

5.2 Area codes

The United States and Canada provide a particularly famous example.

AT&T developed the North American Numbering Plan (NANP) in 1947 to facilitate direct long-distance dialing, with implementation beginning in 1951. (Nanpa)

The basic structure became:

NPA - NXX - XXXX

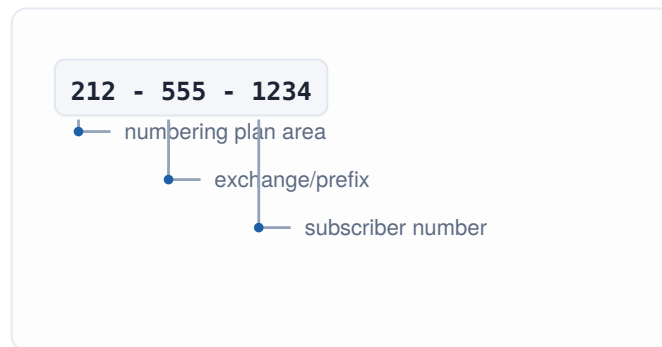


Figure 5.3 Structure of 212 - 555 - 1234

Internationally:

+1 212 555 1234

The 1 identifies the NANP world.

The area code helped the switching system determine the broad destination.

5.3 But numbering isn't necessarily geography anymore

This is another major evolutionary transition.

Originally:

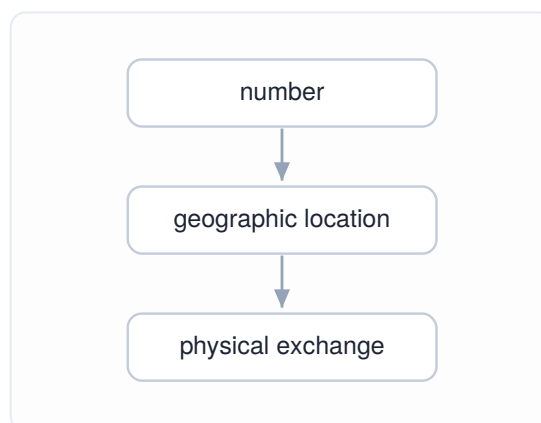


Figure 5.4 number → physical exchange

Modern systems increasingly became:

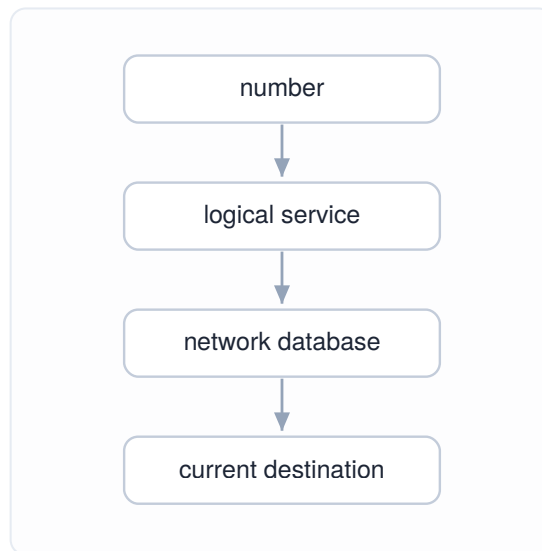


Figure 5.5 number → current destination

This enables:

- mobile numbers
- toll-free numbers
- VoIP
- virtual numbers
- corporate numbering
- unified communications
- number portability
- cloud telephony

The ITU's E.164 framework explicitly covers not only geographic numbering but global services, networks, satellite services and newer IoT/M2M numbering categories. ^{ITU}

5.4 Number portability broke the old model

Suppose:

+91 9876543210

was originally allocated to:

Operator A

You port it:

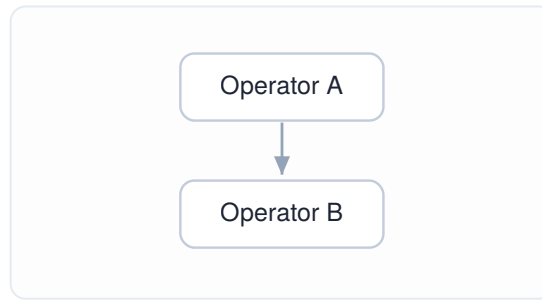


Figure 5.6 Operator A → Operator B

The number doesn't change.

Therefore:

PREFIX

CURRENT OPERATOR

The network needs additional routing intelligence.

A simplified architecture:

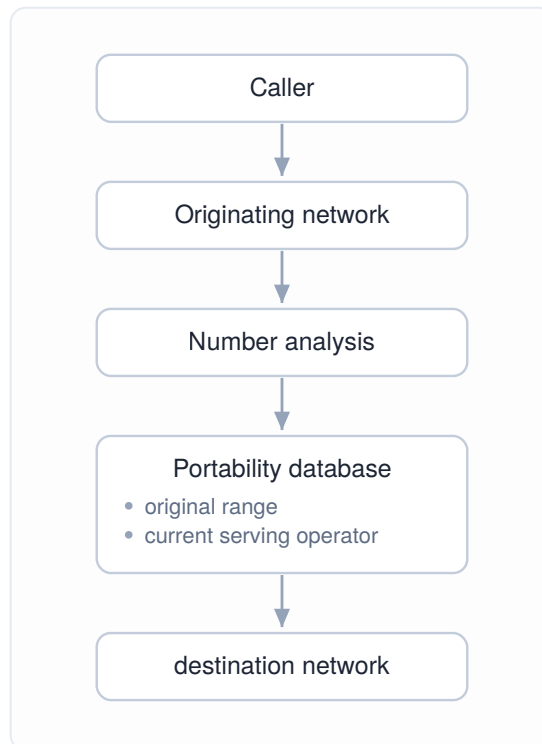


Figure 5.7 Caller → Number analysis → destination network

The FCC's 1996 number portability order explicitly discusses database architecture and routing, illustrating how portability transformed numbering from a purely hierarchical routing mechanism into a database-assisted system. (FCC Documentation)

Mobility: GSM and the Cellular Model

6.1 Mobile networks changed everything

6.2 GSM architecture

6.3 What happens when someone calls your mobile?

6.4 Signaling vs actual voice

6.1 Mobile networks changed everything

Now introduce mobility.

A landline is approximately:

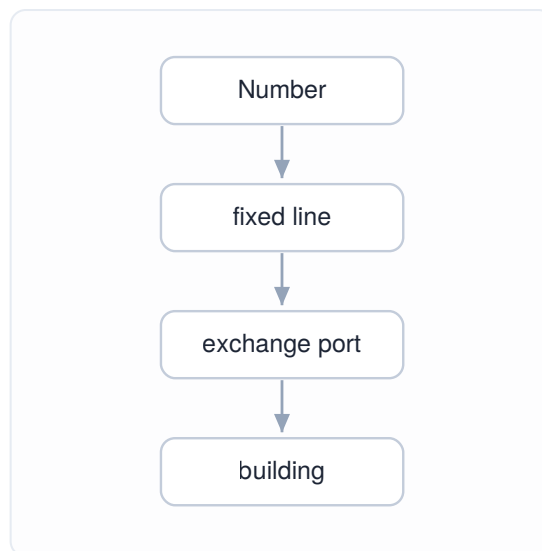


Figure 6.1 Number → building

A mobile phone is:

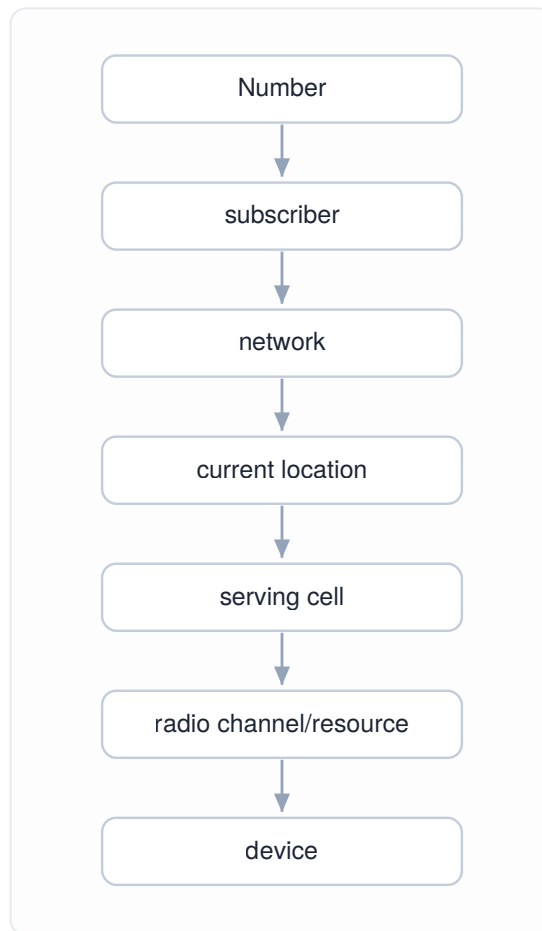


Figure 6.2 Number → current location → device

The subscriber can move:

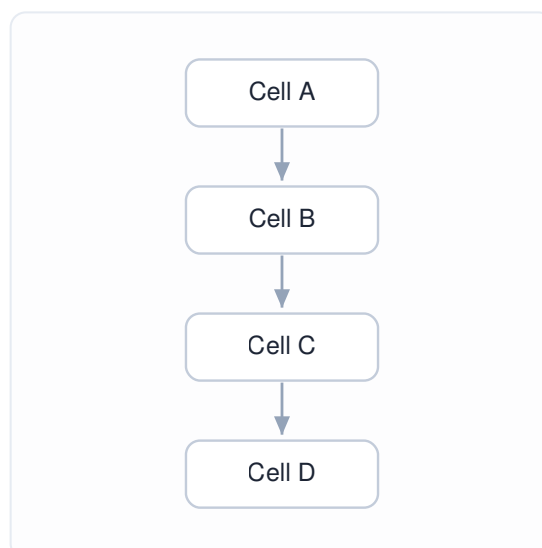


Figure 6.3 Cell A → Cell D

while:

MSISDN = unchanged

IMSI = unchanged

The network therefore needs mobility management.

6.2 GSM architecture

The classic GSM architecture looked approximately like:

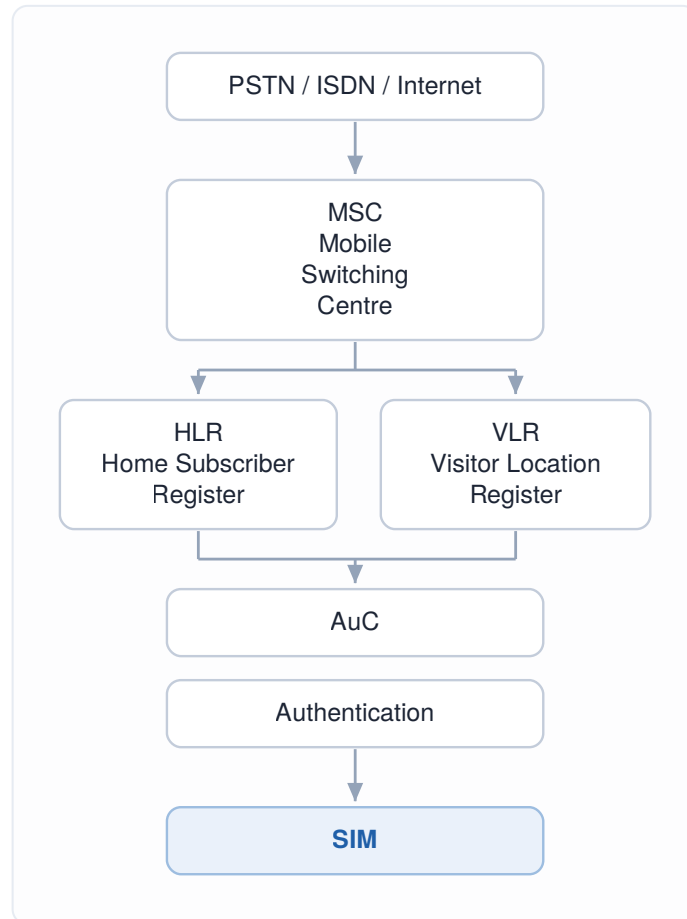


Figure 6.4 PSTN / ISDN / Internet → AuC → SIM

And the radio side:



Figure 6.5 Phone → MSC

So a GSM phone number was only the entry point.

The network then used subscriber and mobility databases to determine where the subscriber currently was.

3GPP/ETSI specifications document subscriber parameters such as IMSI, MSISDN, HLR/VLR information, IMEI and authentication material. (ETSI)

6.3 What happens when someone calls your mobile?

Suppose:



Figure 6.6 Alice → +91 9876543210

A simplified modern cellular process is:

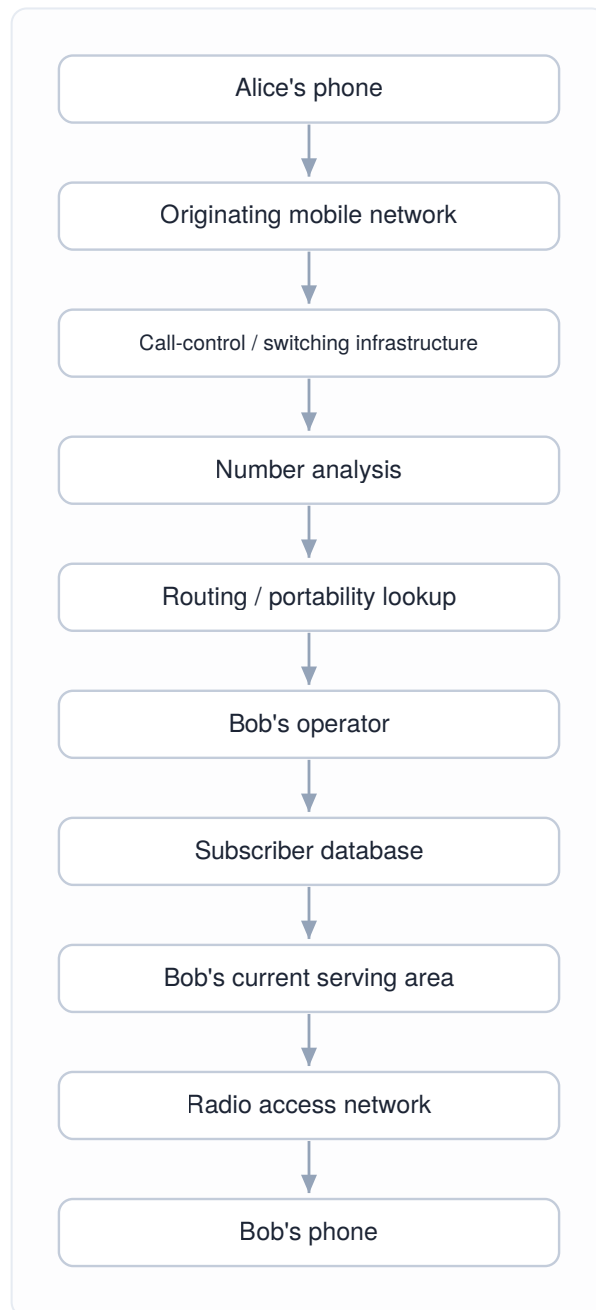


Figure 6.7 Alice's phone → Bob's operator → Bob's phone

The network then establishes the media path.

For traditional circuit-switched voice:

signalling path

voice path

This distinction becomes critical.

6.4 Signaling vs actual voice

This is one of the most important concepts in telephony.

There are essentially two things happening:

- CONTROL PLANE
- "What should happen?"
- DATA/MEDIA PLANE
- "Carry the actual conversation."

Historically:

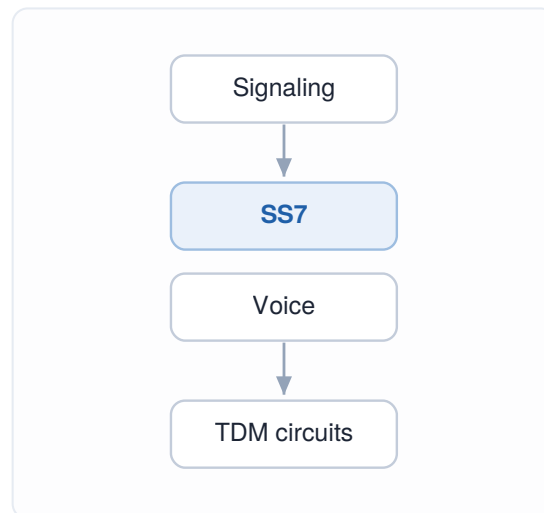


Figure 6.8 Signaling → TDM circuits

Modern systems:

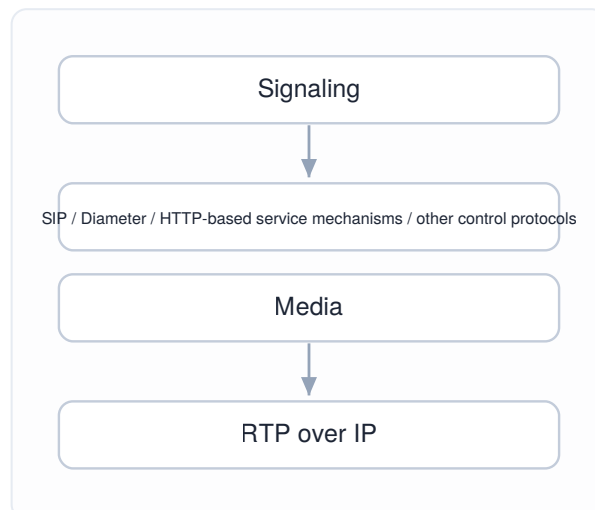


Figure 6.9 Signaling → RTP over IP

So when you dial:

+91 9876543210

the digits don't become the audio stream.

They trigger a chain of control-plane operations that ultimately establish a media session.

Signalling and the Digital PSTN

7.1 SS7

7.3 The telephone number survived the transition

7.2 The PSTN became digital

7.1 SS7

The old PSTN became vastly more sophisticated with digital signaling.

One of the central technologies was:

SS7 — Signaling System No. 7

Instead of using the same voice channel to perform all signaling, networks could use dedicated signaling infrastructure.

Conceptually:

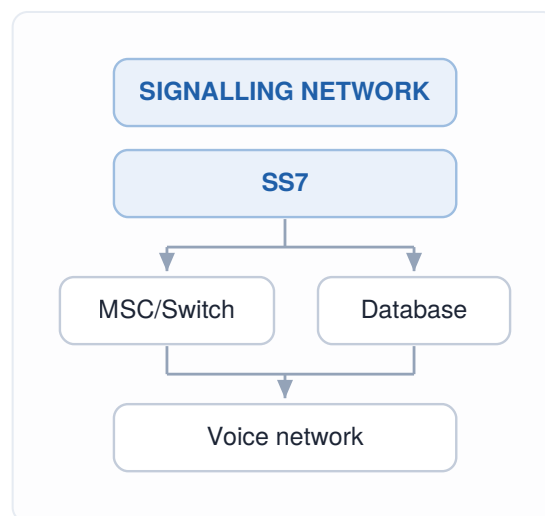


Figure 7.1 Signalling network → Voice network

This allowed things like:

- call setup
- call release
- caller identification
- roaming
- database queries
- number translation

- mobility management
- billing-related signaling

The evolution from electromechanical switching toward electronic switching and SS7 is one of the major transitions in telephone-network architecture. (IEEE Spectrum)

7.2 The PSTN became digital

The old telephone system originally transmitted analog voice.

Then came digital voice.

The basic idea:

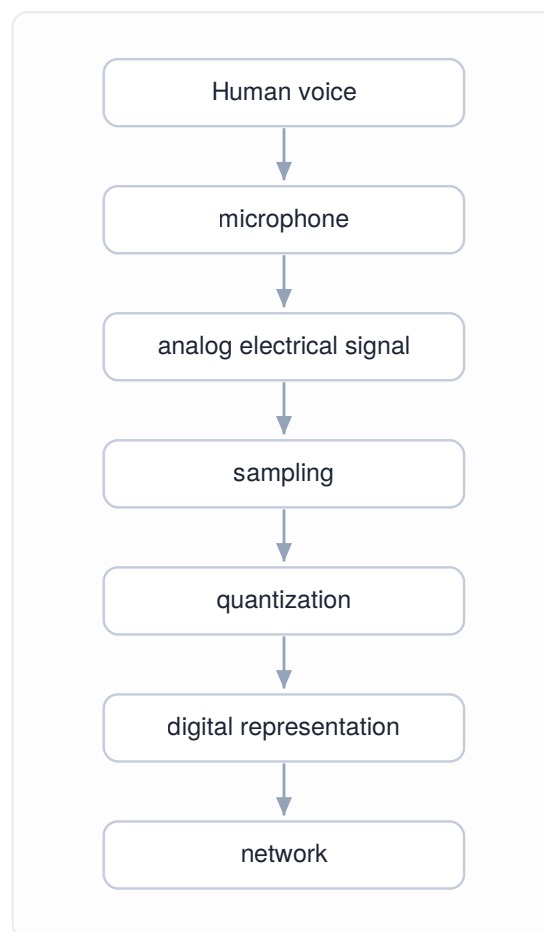


Figure 7.2 Human voice → sampling → network

Classic telephone PCM uses:



Figure 7.3 8 kHz sampling → 64 kbit/s

Hence the famous:

DS0 = 64 kbit/s

Voice channels could then be multiplexed:

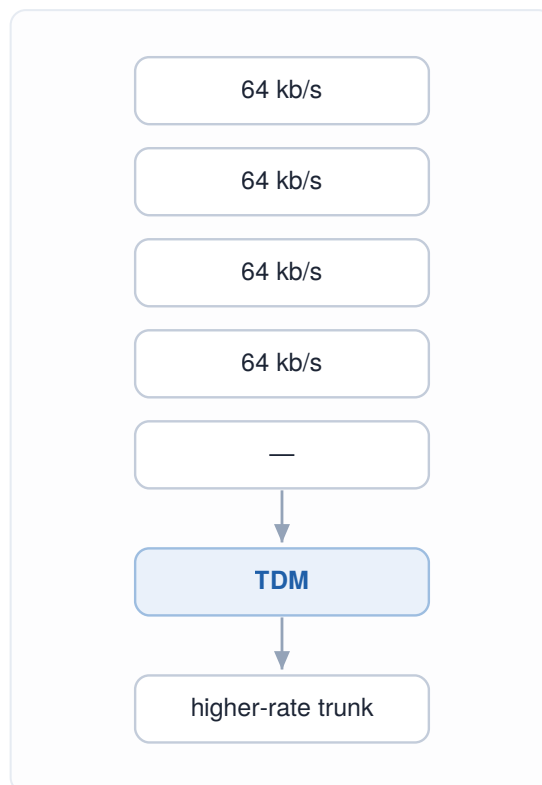


Figure 7.4 64 kb/s → higher-rate trunk

This massively improved network efficiency.

7.3 The telephone number survived the transition

Here's the beautiful engineering hack.

The underlying network could change:

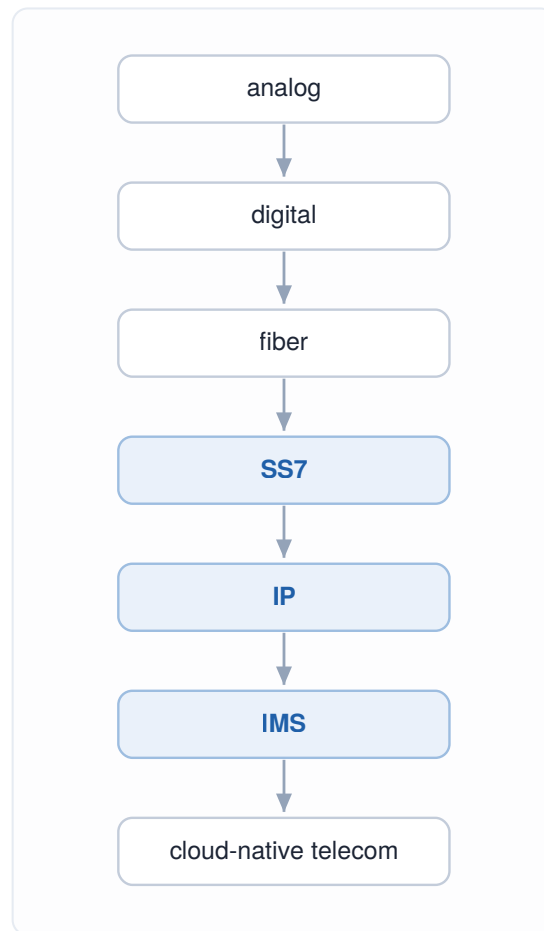


Figure 7.5 analog → SS7 → cloud-native telecom

while the user-facing number remained:

+91 9876543210

That is why numbering systems are so valuable.

They provide application-level stability despite infrastructure-level evolution.

It's analogous to:

DNS name

surviving:

- server migration
- load balancer change
- IP change
- datacenter migration

The identifier persists while the underlying route changes.

The IP Transition: VoIP, SIP and IMS

8.1 VoIP changed the architecture again

8.2 SIP

8.3 Mobile networks eventually moved toward IMS

8.4 The modern stack

8.1 VoIP changed the architecture again

Voice eventually moved onto IP networks.

Instead of:

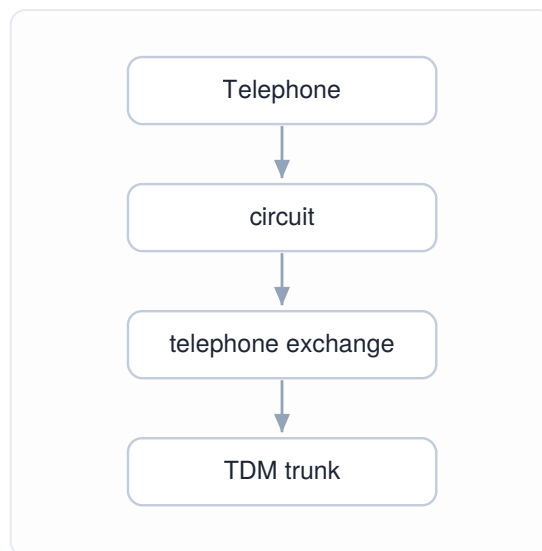


Figure 8.1 Telephone → TDM trunk

we can have:

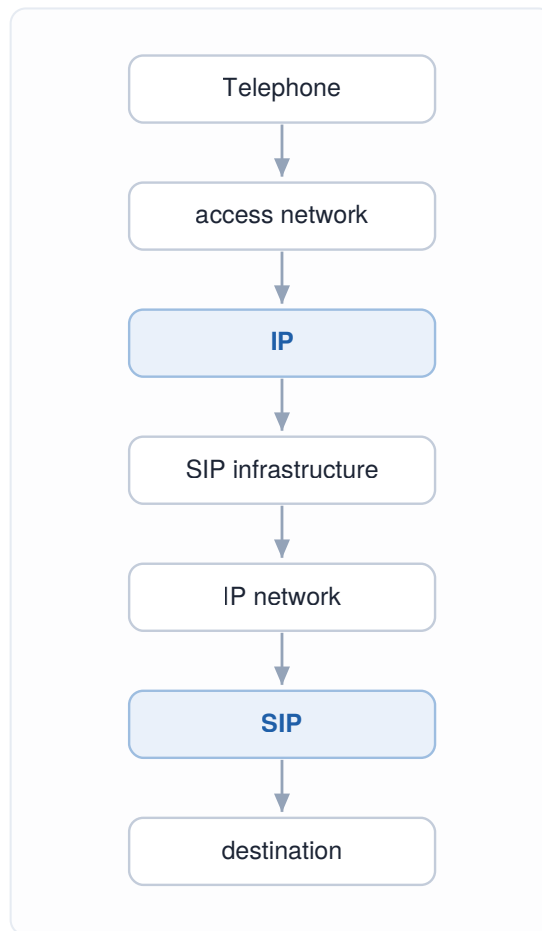


Figure 8.2 Telephone → SIP infrastructure → destination

Now the telephone number can be mapped into an IP-oriented architecture.

This produced:

- VoIP
- SIP phones
- softphones
- hosted PBXs
- cloud telephony
- virtual numbers

8.2 SIP

Modern IP telephony commonly uses SIP, Session Initiation Protocol, for session signaling.

Conceptually:

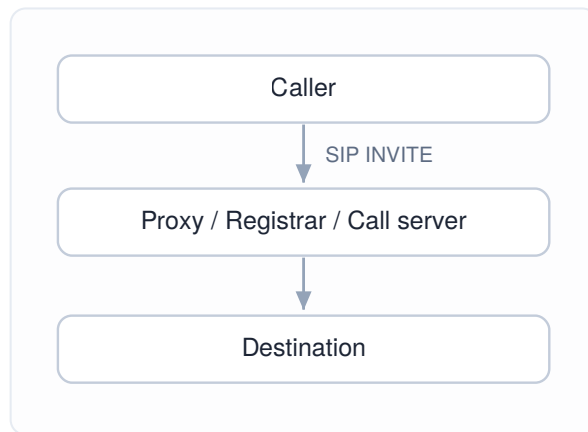


Figure 8.3 Caller → Destination

The actual audio can then be carried using:

RTP

- SIP
- "Let's establish a call."
- RTP
- "Here is the actual audio."

Again:

control plane ≠ media plane

This is the same conceptual separation you see in distributed computing everywhere.

8.3 Mobile networks eventually moved toward IMS

Modern LTE/5G voice architecture is much more IP-centric.

A simplified conceptual architecture:

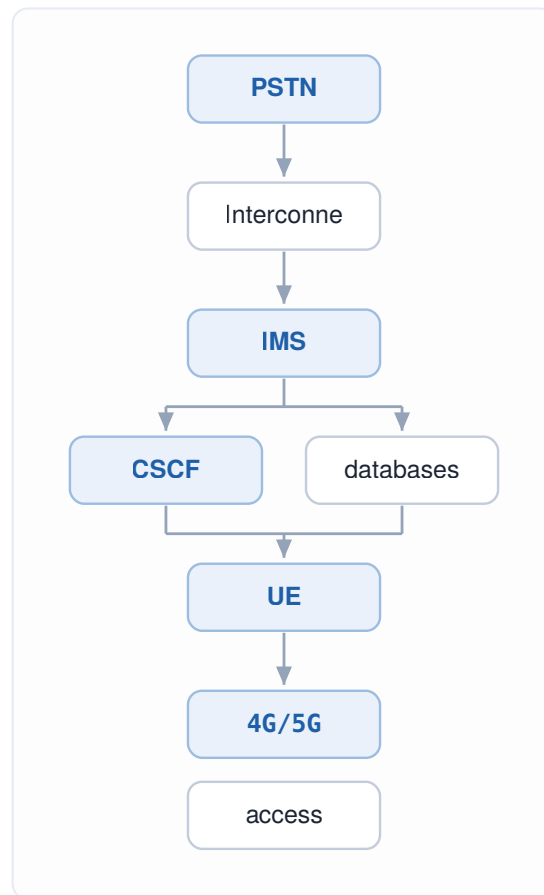


Figure 8.4 PSTN → CSCF / databases → access

IMS means:

IP Multimedia Subsystem

It provides service/control architecture for IP-based telecommunications.

In LTE, voice can be provided through:

VoLTE

In 5G:

VoNR

depending on deployment.

8.4 The modern stack

If you make the abstraction sufficiently deep, a modern phone call can be thought of as:

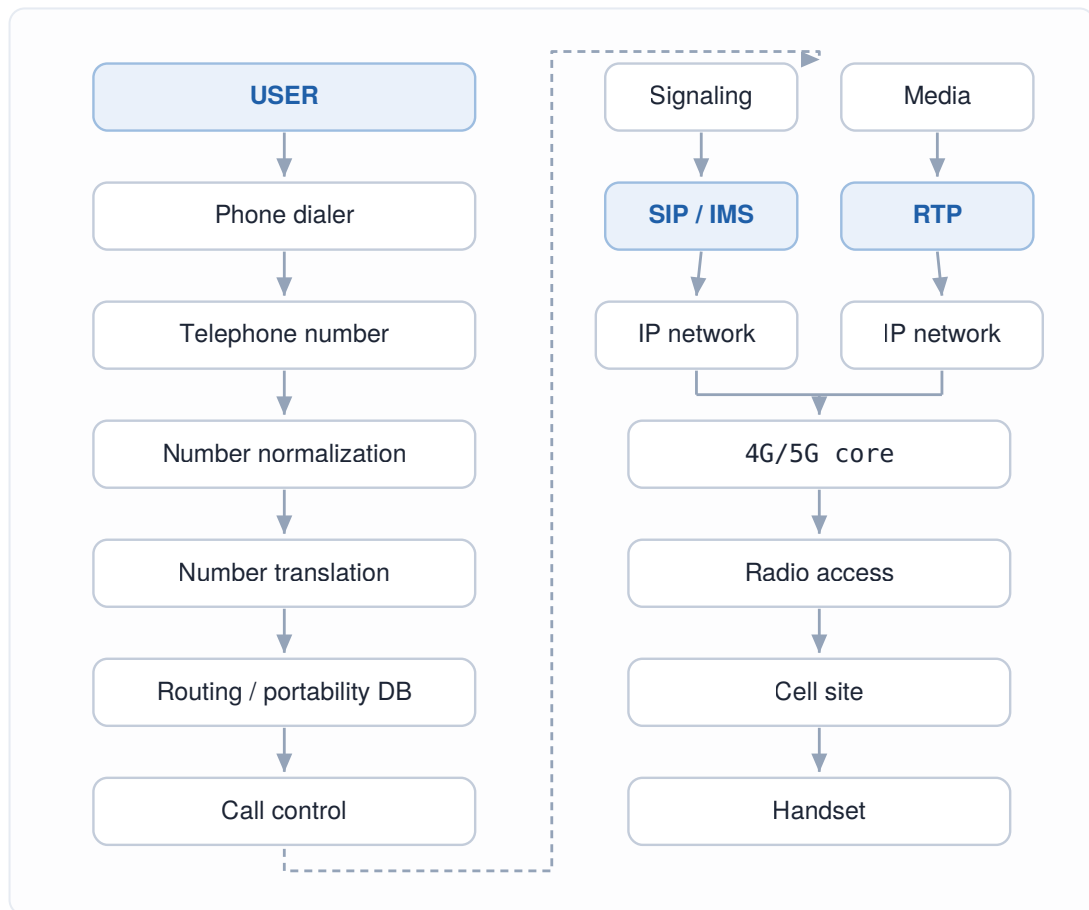


Figure 8.5 USER → Signaling / Media → Handset

That's the conceptual modern architecture.

Radio, Mobility Management and Paging

9.1 The cellular radio side

9.3 Paging

9.2 Why your phone can ring while moving

9.1 The cellular radio side

The telephone number doesn't directly identify:

- tower
- resource block
- frequency
- antenna

Those are dynamically managed.

A modern 5G phone may have:

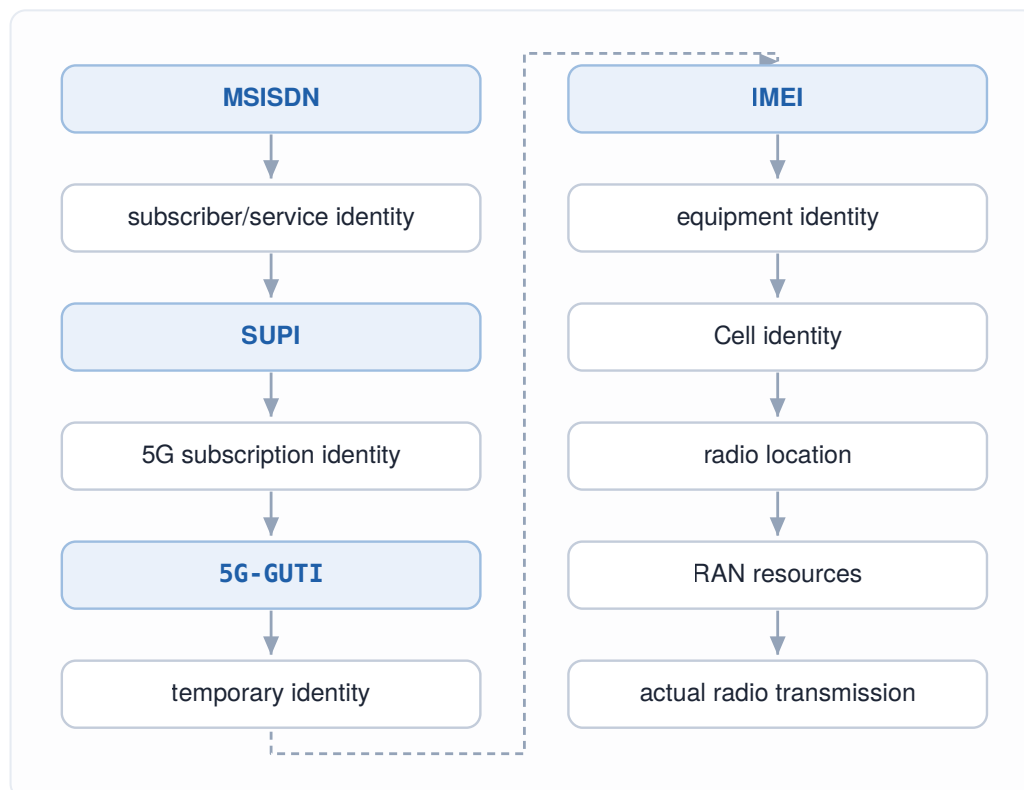


Figure 9.1 MSISDN → IMEI → actual radio transmission

The network deliberately uses temporary identifiers where possible rather than constantly broadcasting permanent subscriber identities.

9.2 Why your phone can ring while moving

Imagine:

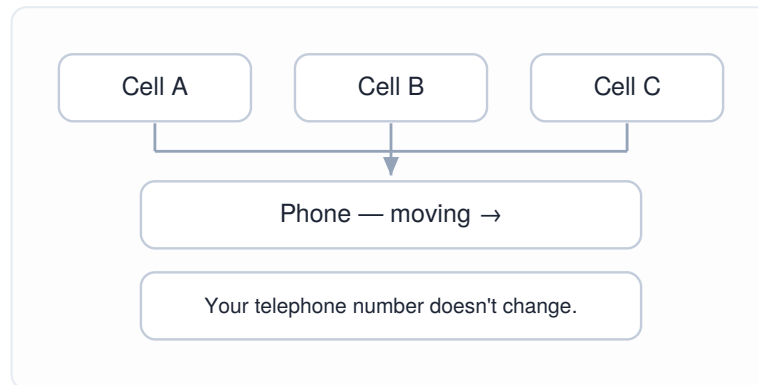


Figure 9.2 Cell A / Cell B / Cell C → Your telephone number doesn't change

Instead, mobility management updates the network's understanding of where your subscription can currently be reached.

Simplified:

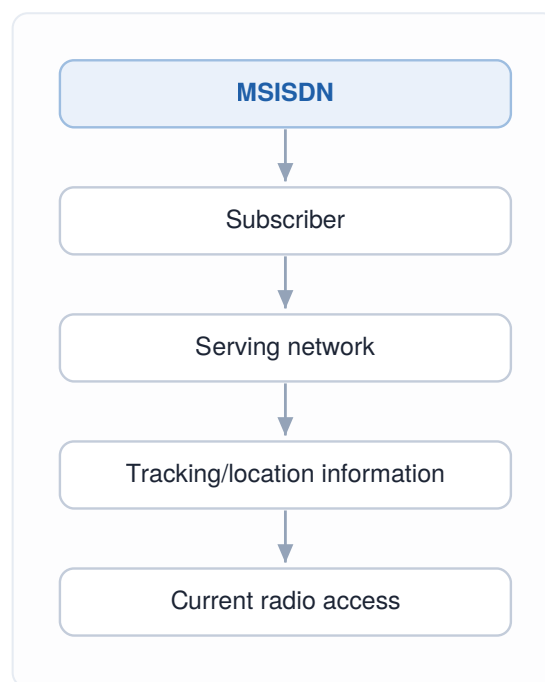


Figure 9.3 MSISDN → Serving network → Current radio access

During mobility, the network performs procedures such as:

- registration
- location updates
- handover
- paging
- authentication
- security context management

So your number is stable while the network attachment state is dynamic.

9.3 Paging

Suppose your phone is idle.

The network doesn't continuously transmit:

"WHERE IS +91 9876543210?"

to every tower on Earth.

That would be an impressively stupid use of infrastructure.

Instead, the network maintains location information at an appropriate granularity.

When a call arrives:

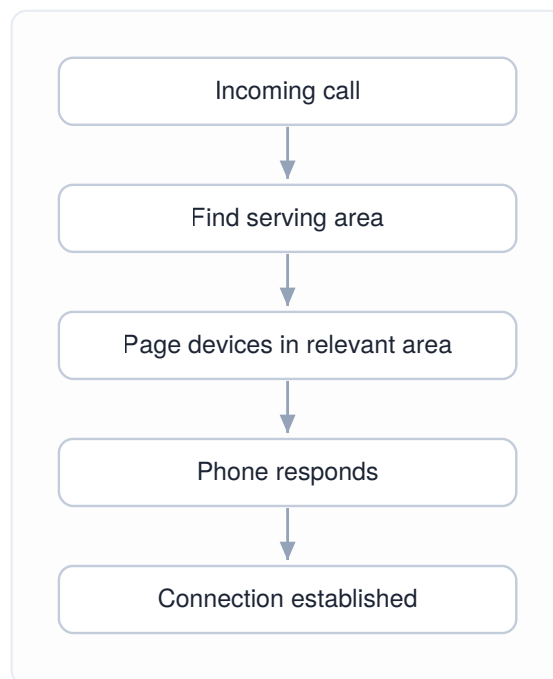


Figure 9.4 Incoming call → Page devices in relevant area → Connection established

This is one reason mobility management is such a fundamental part of cellular architecture.

Databases, Authority and International Routing

10.1 Numbering databases

10.2 Who controls a phone number?

10.3 What happens when you call another country?

10.1 Numbering databases

Modern telecommunications heavily depends on databases.

Conceptually:

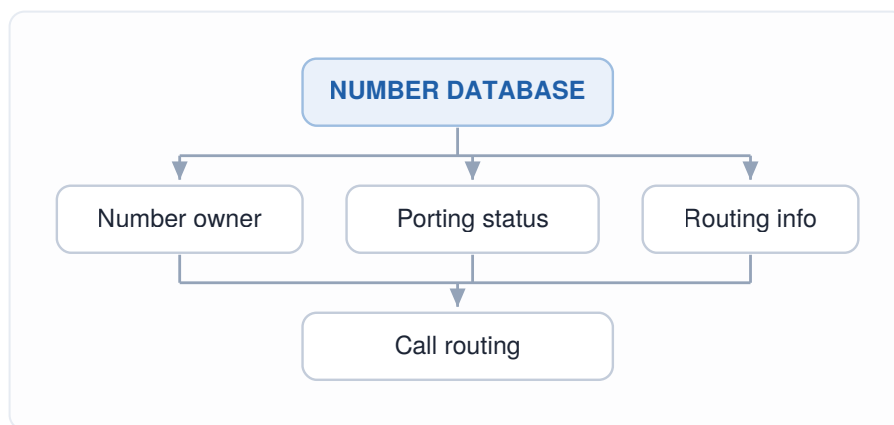


Figure 10.1 Number database → Call routing

The exact databases and protocols vary enormously between countries and technologies.

There is no single magical:

GLOBAL_PHONE_NUMBER_DATABASE

Instead there are:

- international numbering administration
- national numbering authorities
- operator databases
- portability databases
- subscriber databases
- routing databases
- service databases

This is a distributed administrative and technical ecosystem.

10.2 Who controls a phone number?

This is another layer people often misunderstand.

There are several levels:

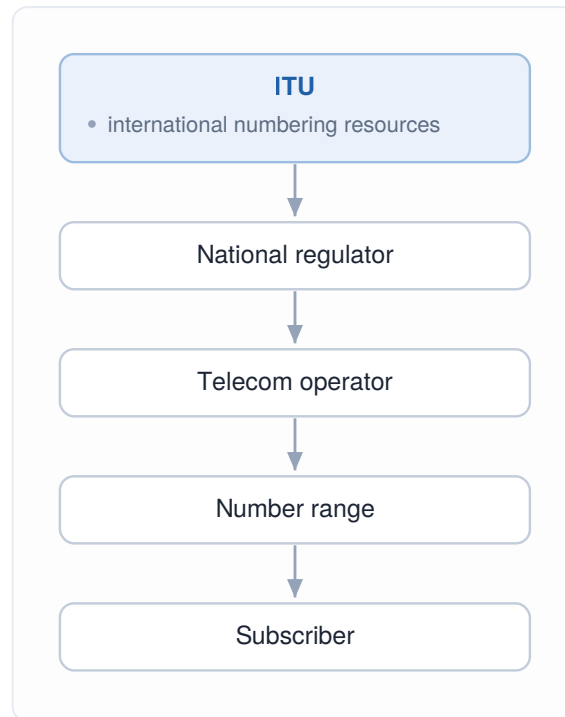


Figure 10.2 ITU → Telecom operator → Subscriber

The exact administrative chain varies by country.

In India, numbering resources are administered within the national telecom regulatory/government framework, while operators receive numbering resources for services.

The number isn't literally "owned" by you in the same sense that you own a physical object.

You have a right to use it under the applicable telecommunications service arrangement.

10.3 What happens when you call another country?

Suppose:

- | | | |
|------------------|---------|-------------------|
| • India | • calls | • +1 212 555 1234 |
| • +91 9876543210 | • USA | |

Conceptually:

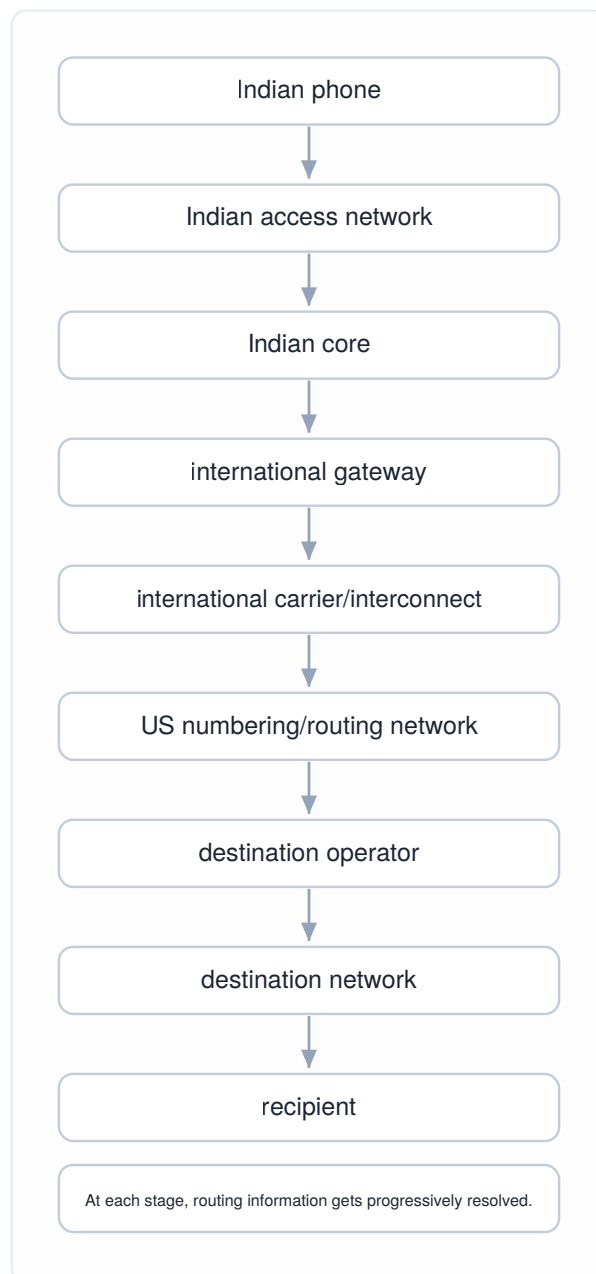


Figure 10.3 Indian phone → US numbering/routing network → At each stage, routing information gets progressively resolved

This is essentially hierarchical distributed routing.

Comparisons and Boundary Cases

11.1 The relationship with DNS

11.2 Phone number vs IP address

11.3 Phone number vs SIM

11.4 One person can have multiple numbers

11.1 The relationship with DNS

There is a fascinating analogy:

DNS

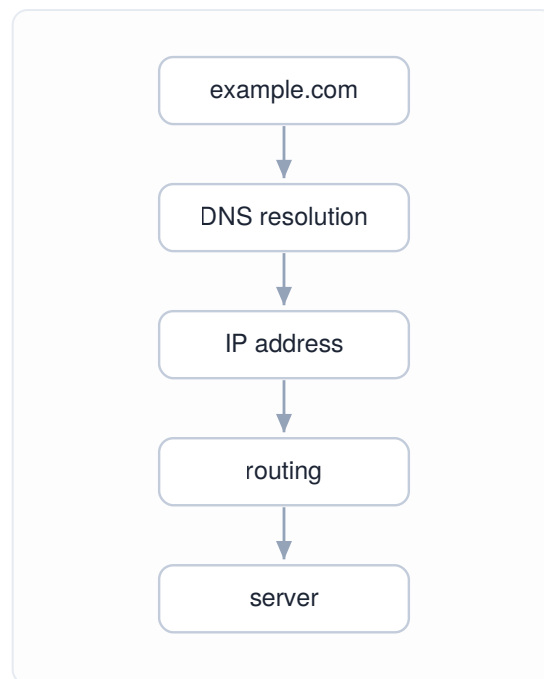


Figure 11.1 example.com → IP address → server

Telephony

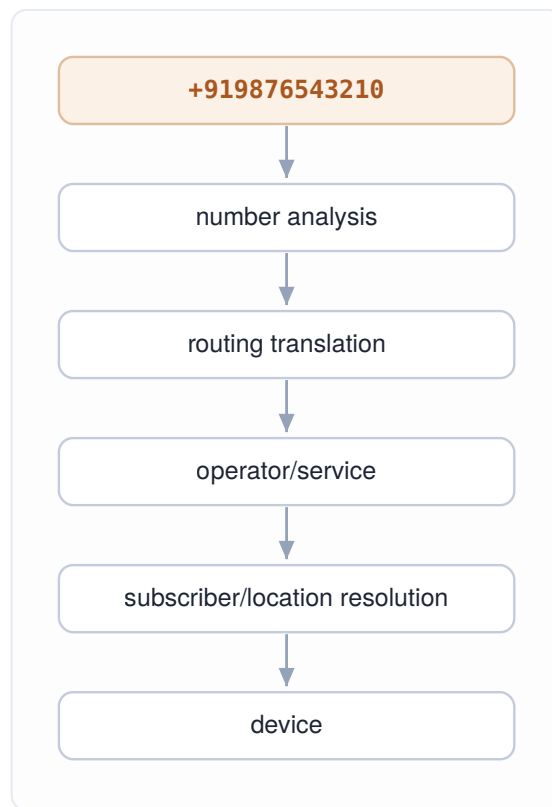


Figure 11.2 +919876543210 → operator/service → device

Both are forms of:

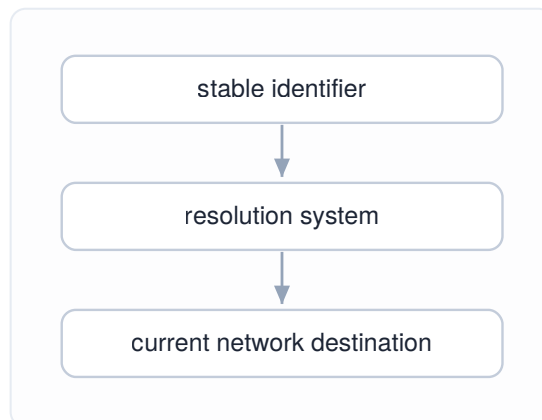


Figure 11.3 stable identifier → current network destination

But telephone numbering predates the Internet by roughly a century.

11.2 Phone number vs IP address

Do not think:

phone number = IP address

They operate at different abstraction layers.

For example:

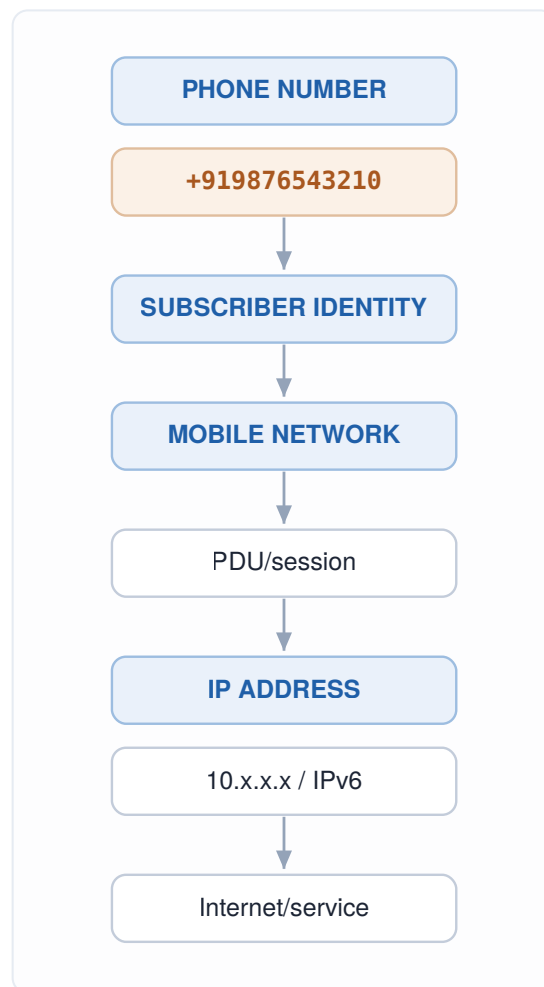


Figure 11.4 Phone number → PDU/session → Internet/service

Your phone can have one telephone number while its packet-network address changes.

11.3 Phone number vs SIM

Likewise:

phone number $\neq$ SIM

You can replace your SIM while retaining your number through provisioning/porting mechanisms.

And:

SIM $\neq$ phone

You can move the subscription between devices.

And:

phone number $\neq$ IMEI

The IMEI identifies equipment.

So a simplified identity graph is:

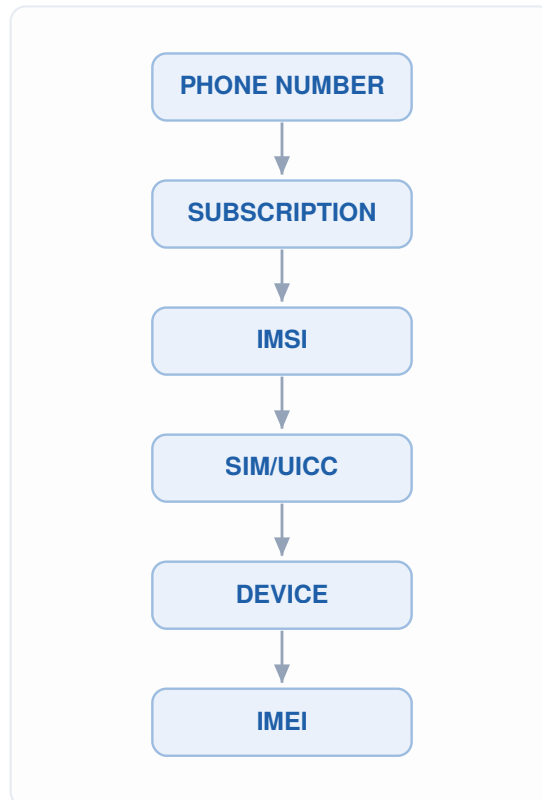


Figure 11.5 Phone number → SIM/UICC → IMEI

But real-world systems can have multiple numbers, subscriptions and identities associated with one device or account.

11.4 One person can have multiple numbers

There is no fundamental rule:

human = phone number

You can have:

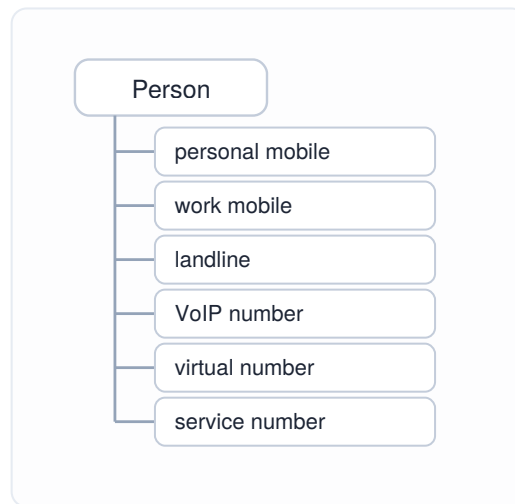


Figure 11.6 Person and its constituent parts

Likewise:

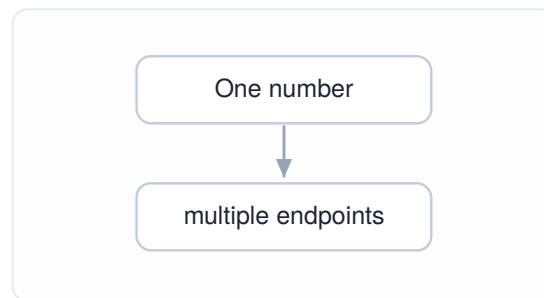


Figure 11.7 One number → multiple endpoints

can exist in business telephony.

For example:

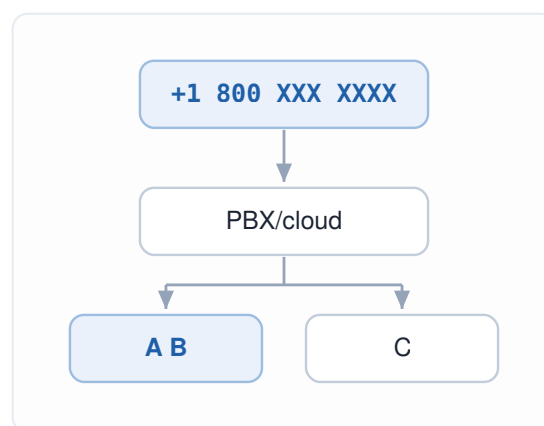


Figure 11.8 +1 800 xxx xxxx → A B / C

So a phone number can identify a service, not necessarily a single physical device.

Special Numbers and Overlaid Services

12.1 Emergency numbers

12.2 Toll-free numbers

12.3 Voicemail

12.4 Caller ID

12.5 SMS uses the same numbering namespace

12.6 WhatsApp demonstrates another layer

12.1 Emergency numbers

Some numbers don't follow normal subscriber numbering.

For example:

112
911
999

These are service identifiers.

The network recognizes them specially:

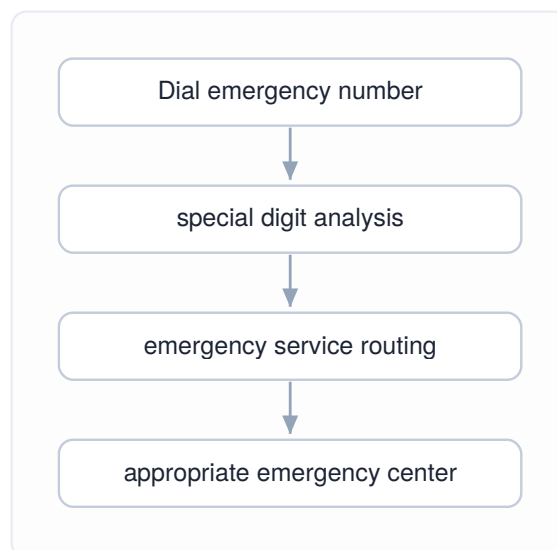


Figure 12.1 Dial emergency number → appropriate emergency center

They therefore demonstrate that the telephone numbering system isn't simply:

number → subscriber

It can be:

number → service

12.2 Toll-free numbers

Similarly:

800...

doesn't necessarily mean:

physical phone at location X

Instead:

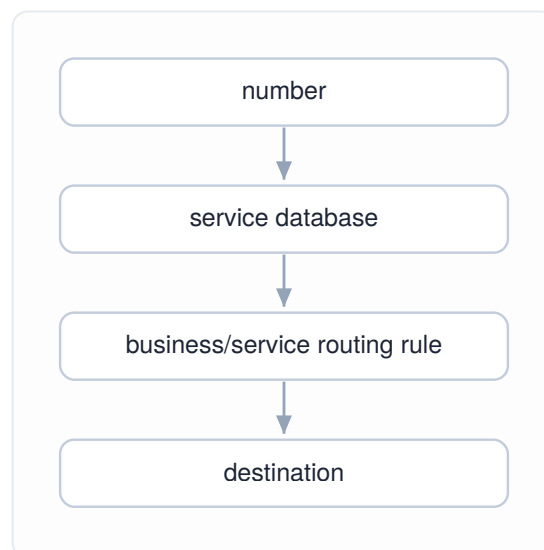


Figure 12.2 number → destination

The network can determine:

- time
- location
- originating carrier
- load
- destination availability

and select an appropriate endpoint.

Telephone numbers therefore evolved into a programmable routing namespace.

12.3 Voicemail

Consider:

+91 9876543210

You call it.

If unavailable:

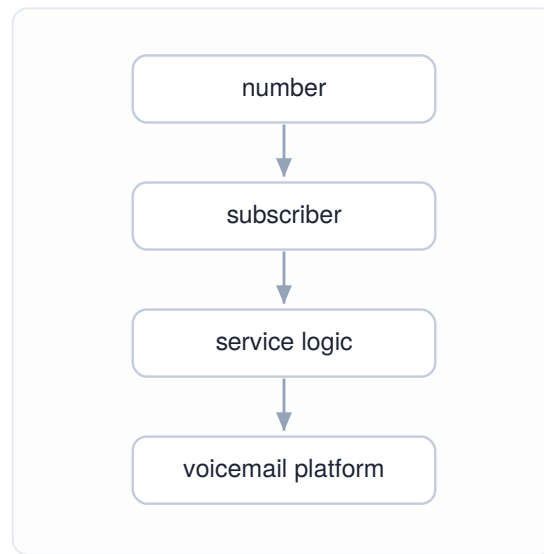


Figure 12.3 number → voicemail platform

The number itself didn't change.

The network/service logic determined:

"this subscriber is unavailable"

and routed the session to another service endpoint.

12.4 Caller ID

When you call someone, their phone can display:

+91 9876543210

That information is transmitted through signaling mechanisms.

It is not the phone magically discovering your number.

Conceptually:

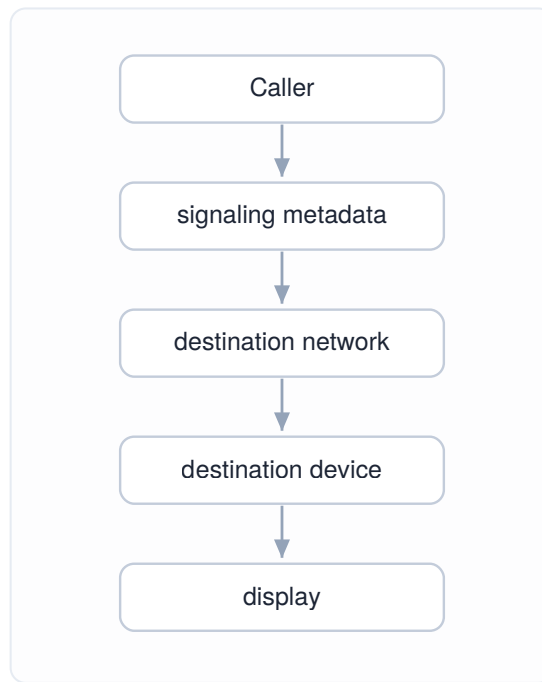


Figure 12.4 Caller → destination network → display

And because signaling information can be manipulated under certain circumstances, caller ID is not cryptographic proof of identity.

That is why spoofing exists.

12.5 SMS uses the same numbering namespace

SMS historically uses telephone numbering as an addressing mechanism.

Simplified:

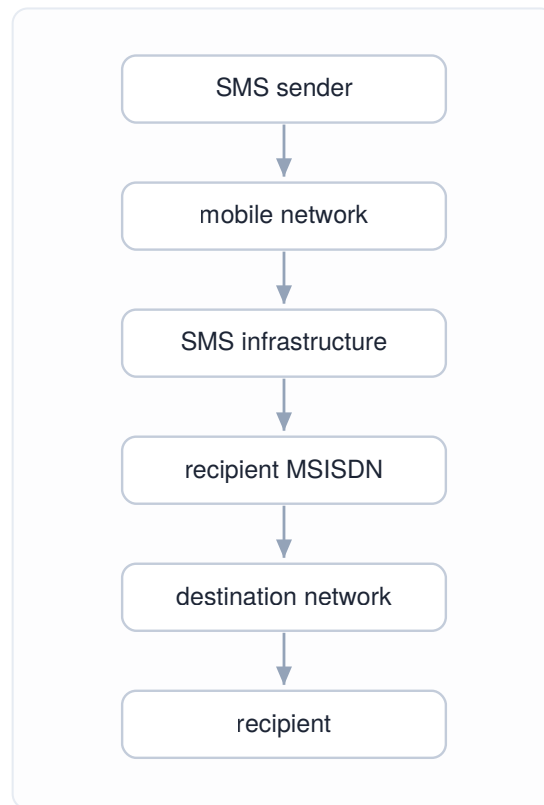


Figure 12.5 SMS sender → recipient MSISDN → recipient

The SMS infrastructure determines where to deliver the message using subscriber/network information.

phone number

became useful beyond voice.

12.6 WhatsApp demonstrates another layer

Suppose WhatsApp registers:

+91 9876543210

WhatsApp uses the telephone number as an account identifier / bootstrap identity.

But after registration, the actual communication doesn't mean:



Figure 12.6 WhatsApp → phone number

Instead:

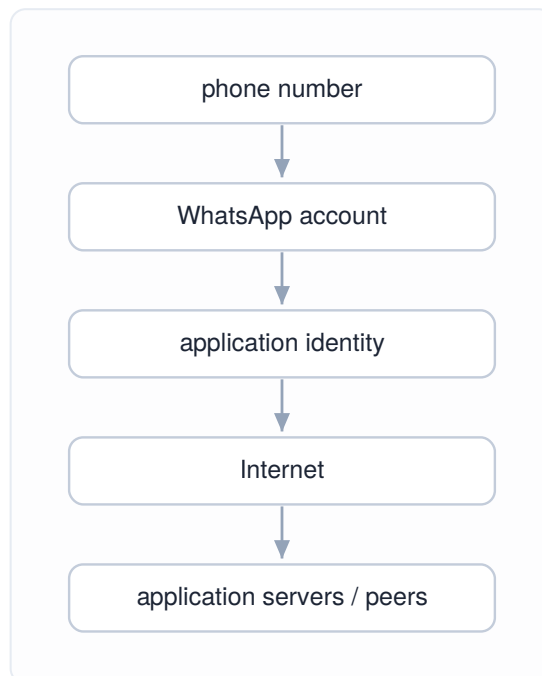


Figure 12.7 phone number → application identity → application servers / peers

This is another example of:

A telephone number can be used as an identifier without the telephone network carrying the actual communication.

The Whole Evolution in One View

13.1 The entire historical evolution

13.2 The most important conceptual transformation

13.3 The modern conceptual model

13.4 The really deep abstraction

13.1 The entire historical evolution

You can compress the history into this:

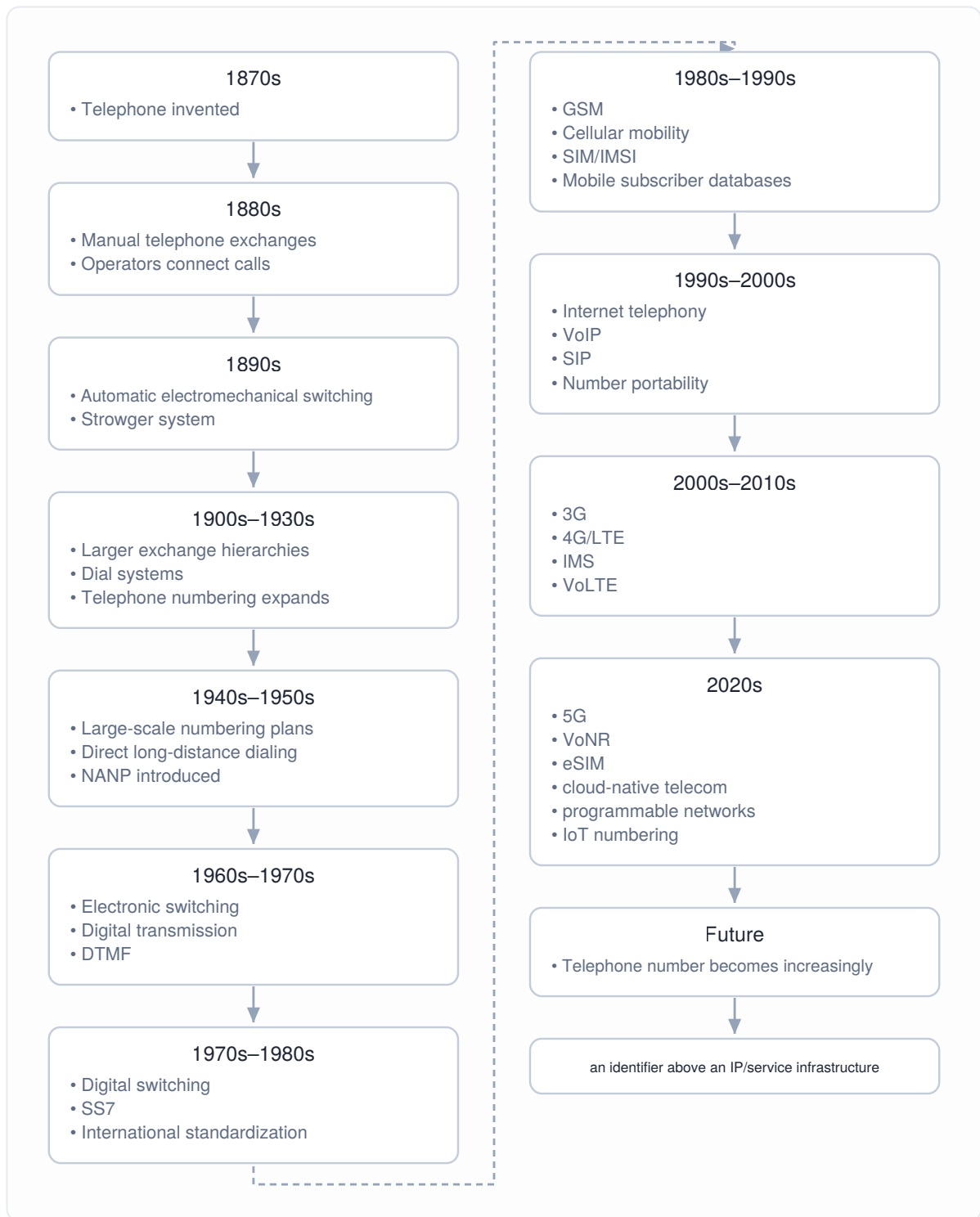


Figure 13.1 1870s → 1970s–1980s → an identifier above an IP/service infrastructure

The key historical pattern is:

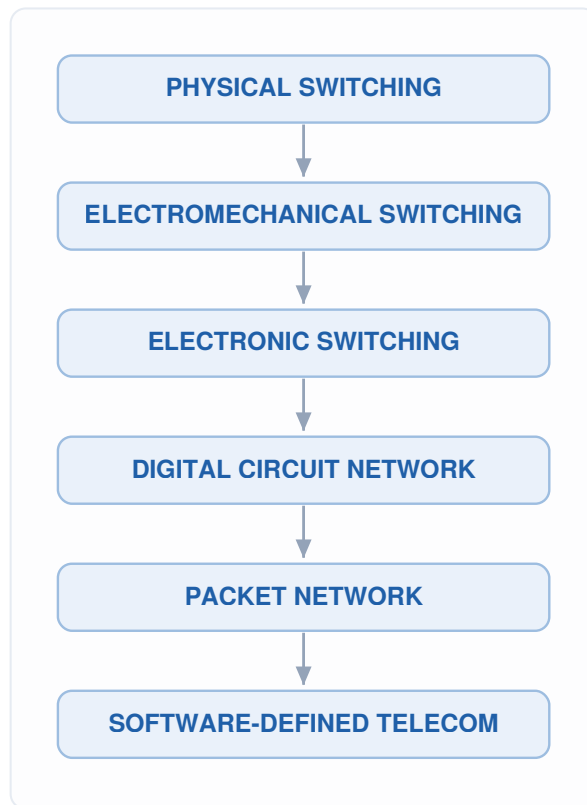


Figure 13.2 Physical switching → Digital circuit network → Software-defined telecom

while:

telephone number

survived almost all of it.

13.2 The most important conceptual transformation

Originally:

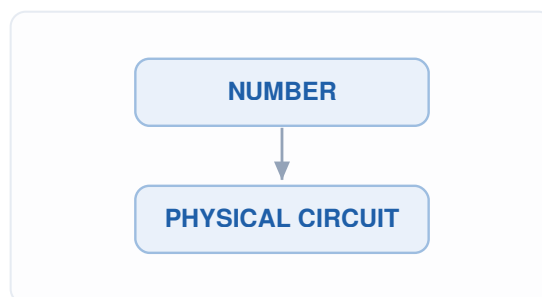


Figure 13.3 NUMBER → Physical circuit

Then:

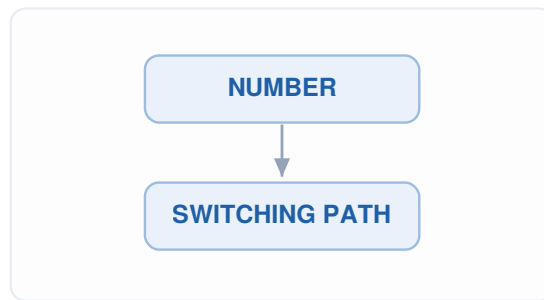


Figure 13.4 NUMBER → Switching path

Then:

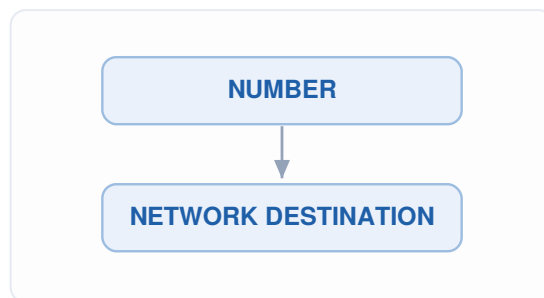


Figure 13.5 NUMBER → Network destination

Then:

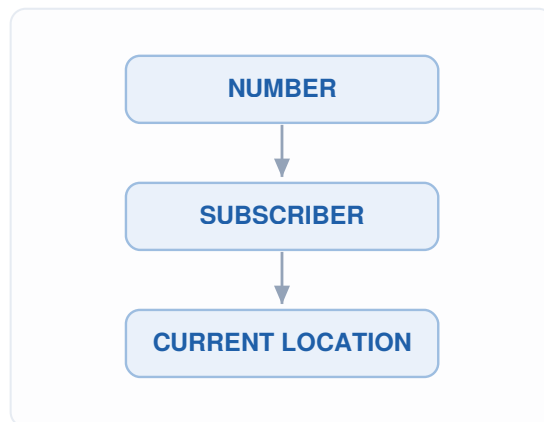


Figure 13.6 NUMBER → Current location

And increasingly:

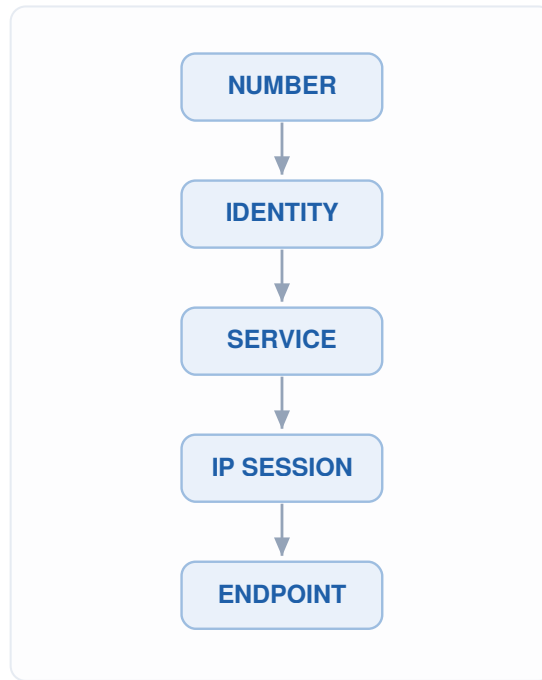


Figure 13.7 NUMBER → SERVICE → ENDPOINT

That is the actual evolution.

13.3 The modern conceptual model

If you want the whole thing represented as one architecture diagram:

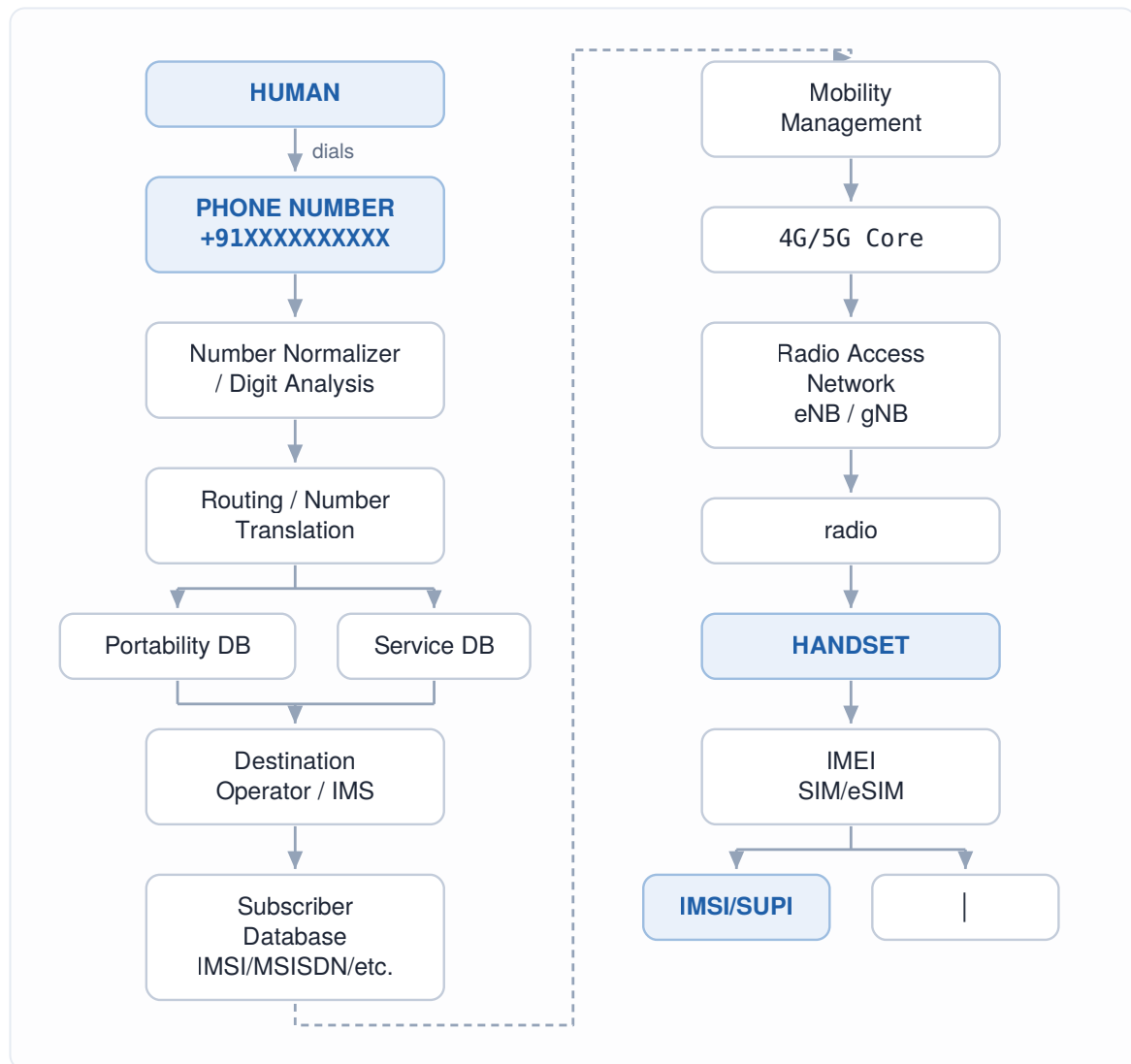


Figure 13.8 HUMAN → Mobility → Imsi/supi / |

And alongside that:

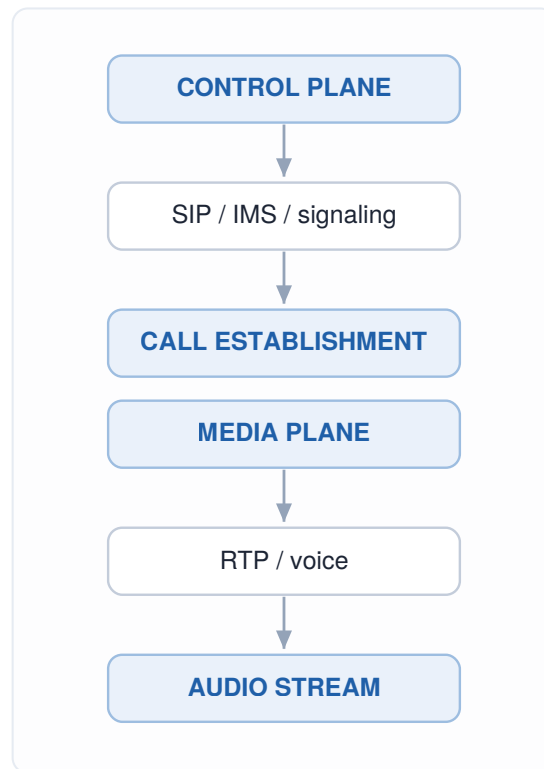


Figure 13.9 Control plane → Media plane → Audio stream

13.4 The really deep abstraction

Since you're studying computer architecture and systems, this is probably the most useful way to think about it.

A telephone number is essentially an indirection layer.

Compare:

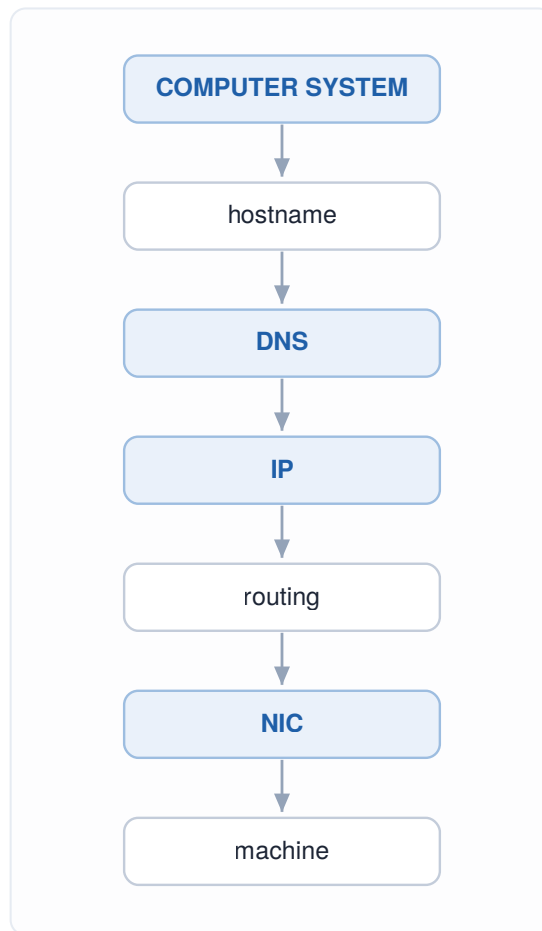


Figure 13.10 Computer system → IP → machine

with:

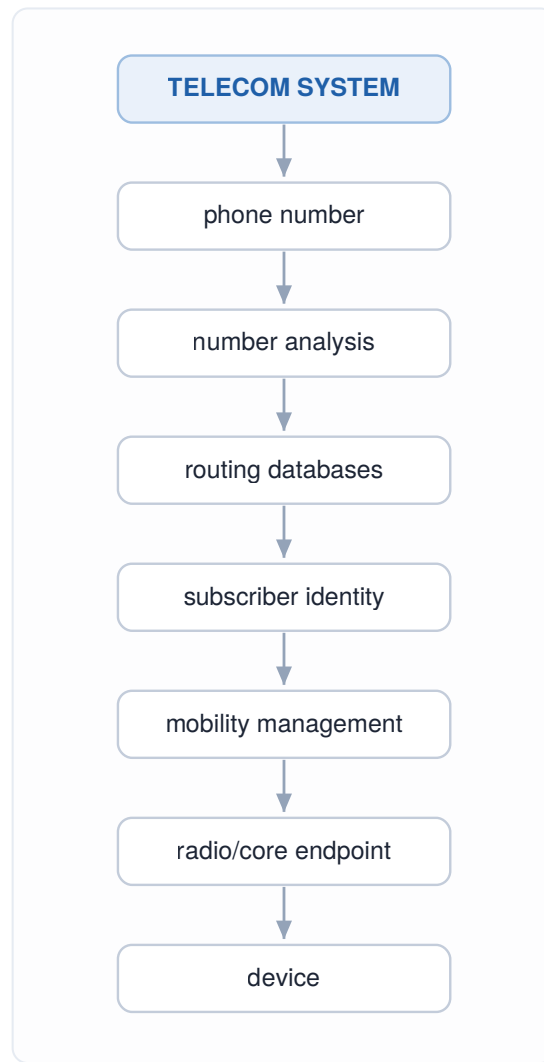


Figure 13.11 Telecom system → subscriber identity → device

The telephone network spent more than a century gradually moving the mapping:

number → physical circuit

toward:

number → logical identity → dynamic endpoint

That is why the number itself has remained remarkably stable even though practically every layer underneath it has been replaced.

And that is also why you should not memorize:

"The first four digits tell you the operator."

That is an artifact of a particular numbering/allocation regime, not the fundamental architecture. Number portability exists specifically because the logical identifier and underlying network destination are separable.

The hierarchy to remember

If you want one mental model that survives almost every telecommunication technology:

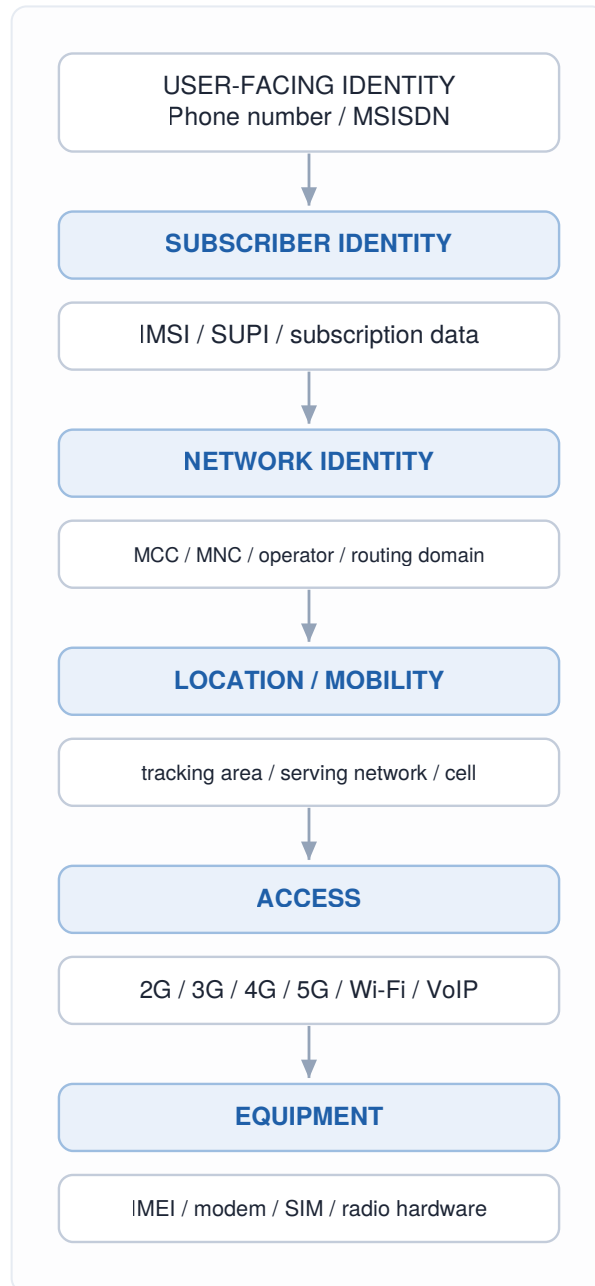


Figure 13.12 User-facing identity → Location / mobility → IMEI / modem / SIM / radio hardware

That's the architecture underneath the innocent little 10 digits humans type into a contact list.

The standards themselves are still evolving: the ITU issued the current E.164 revision in February 2026, including numbering provisions for newer categories such as IoT/M2M. ^{ITU}

Primary technical references

Primary technical references

- ITU-T E.164, International Public Telecommunication Numbering Plan
- ITU-T E.164.1, allocation and reclamation of country codes
- NANPA, history and structure of the North American Numbering Plan
- TRAI, Mobile Number Portability framework
- ETSI/3GPP TS 23.008, mobile subscriber data architecture
- GSMA mobile identity glossary

The natural next layer to study is exactly what happens inside the network after you press the Call button, down to MSISDN → IMSI/SUPI → HLR/HSS/UDM → MSC/IMS → SS7/Diameter/SIP → gNB → paging → bearer establishment → RTP voice. That is where the phone-number abstraction finally collides with actual telecom machinery.